
Geometrical optics

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Your reflection in the mirror, a straw in a glass of water, telescopes, communications, spotlights and car headlights. All these things rely on the way light reflects off surfaces or refracts in different media. If light did not reflect and refract as it does, you would not be able to see how you look before going out or communicate over long distances.

Key Mathematics Concepts

- Equations — Mathematics, Grade 10, Equations and inequalities
- Trigonometry — Mathematics, Grade 10, Trigonometry

5.1 Summary of properties of light

ESBMW

Imagine you are indoors on a sunny day. A beam of sunlight through a window lights up a section of the floor. How would you draw this sunbeam? You might draw a series of parallel lines showing the path of the sunlight from the window to the floor. This is not exactly accurate — no matter how hard you look, you will not find unique lines of light in the sunbeam! However, this is a good way to draw light and to model light geometrically, as we will see in this chapter.

We call these narrow, imaginary lines of light **light rays**. Recall that light can behave like a wave and so you can think of a light ray as the path of a point on the crest of a wave.

We can use light rays to model the behaviour of light relative to mirrors, lenses, telescopes, microscopes, and prisms. The study of how light interacts with materials is called **optics**. When dealing with light rays, we are usually interested in the shape of a material and the angles at which light rays hit it. From these angles, we can determine, for example, the distance between an object and its reflection. We call these methods **geometrical optics**.



5.2 Light rays

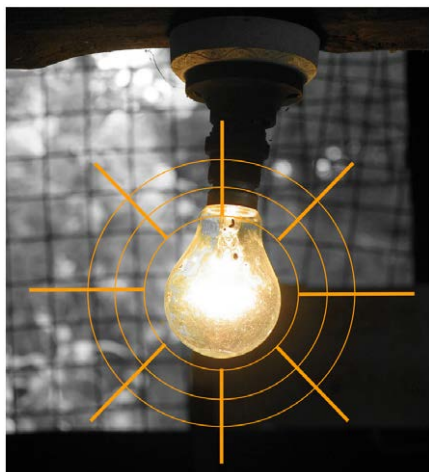
ESBMX

In physics we use the idea of a *light ray* to indicate the direction in which light travels. In geometrical optics, we represent light rays with straight arrows to show how light

propagates. Light rays are not an exact description of the light; rather they show the direction in which the light wavefronts are travelling.

DEFINITION: *Light ray*

Light rays are lines which are perpendicular to the light's wavefronts. In geometrical optics we represent light rays with arrows with straight lines.



The lightbulb is a source of light. The light wavefronts are shown by the concentric circles coming from the bulb. We represent the direction in which the wavefronts are moving by drawing light rays (the arrows) perpendicular to the wavefronts.

In Figure 5.1, the light rays from the object enter the eye and the eye sees the object. Note that we can *only* see an object when light from the object enters our eyes. The object must be a source of light (e.g. a light bulb) or else it must reflect light from a source (e.g. the moon which reflects light from the sun), and the light must enter our eyes.

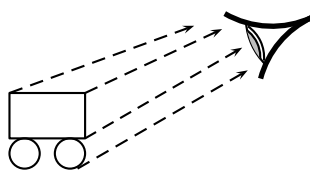


Figure 5.1: We can only see an object when light from that object enters our eyes. When the light travels from the object to the eye, the eye can see the object. Light rays entering the eye from the cart are shown as dashed lines. The second wheel of the cart will be invisible as no straight, unobstructed lines exist between it and the eye.

From the figures you can see that the light rays showing the path of light are straight arrows. In geometrical optics, light travels in straight lines.

Investigation: To demonstrate that light travels in straight lines

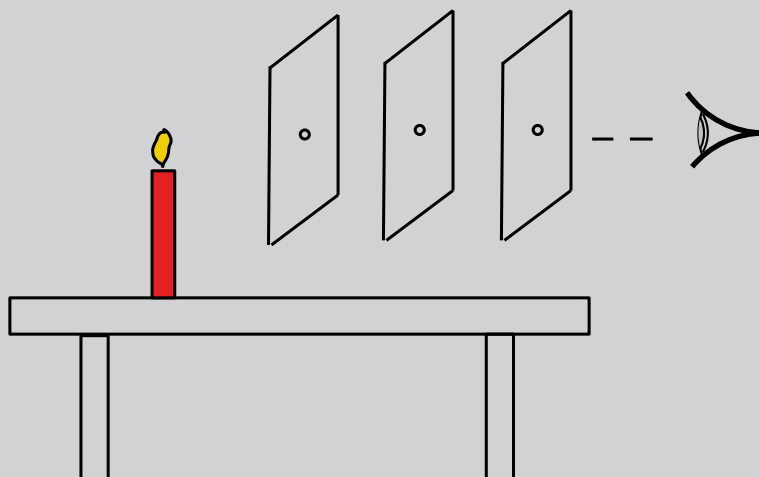
Apparatus:

You will need a candle, matches and three sheets of paper.

Method:

1. Make a small hole in the middle of each of the three sheets of paper.
2. Light the candle.

3. Look at the burning candle through the hole in the first sheet of paper.
4. Place the second sheet of paper between you and the candle so that you can still see the candle through the holes.
5. Now do the same with the third sheet so that you can still see the candle. The sheets of paper must not touch each other.



6. What do you notice about the holes in the paper?

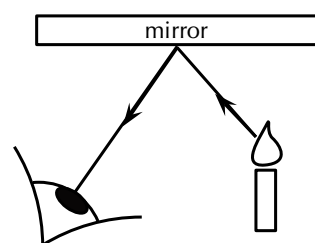
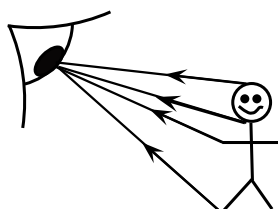
Conclusions:

In the investigation you will notice that the holes in the paper need to be in a straight line. This shows that light travels in a straight line. We cannot see around corners. This also proves that light does not bend around a corner, but travels straight.

Ray diagrams

ESBMV

A ray diagram is a drawing that shows the path of light rays. Light rays are drawn using straight lines and arrow heads. The figure below shows some examples of ray diagrams.



5.3 Properties of light: revision

ESBMZ

When light interacts with objects or a medium, such as glass or water, it displays certain properties: it can either be **reflected**, **absorbed** or **transmitted**.

When you smile into a mirror, you see your own face smiling back at you. This is caused by the reflection of light rays on the mirror. Reflection occurs when an incoming light ray bounces off a surface. In the following figure, a still lake reflects the landscape surrounding it.

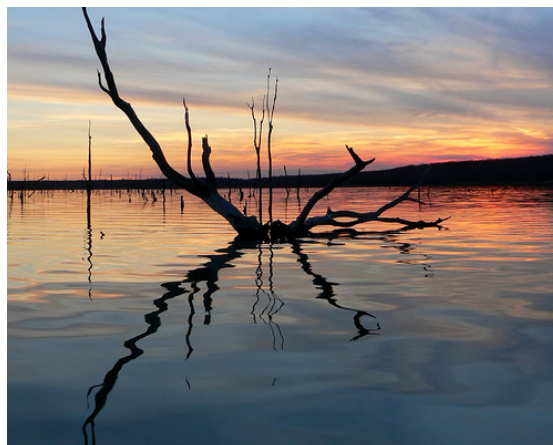


Figure 5.2: A landscape reflection from a still lake.

To describe the reflection of light, we will use the following terminology. The incoming light ray is called the **incident ray**. The light ray moving away from the surface is the **reflected ray**. The most important characteristic of these rays is their angles in relation to the reflecting surface. These angles are measured with respect to the normal of the surface. The **normal** is an imaginary line perpendicular to the surface. The **angle of incidence**, θ_i is measured between the incident ray and the surface normal. The **angle of reflection**, θ_r is measured between the reflected ray and the surface normal. This is shown in Figure 5.3.

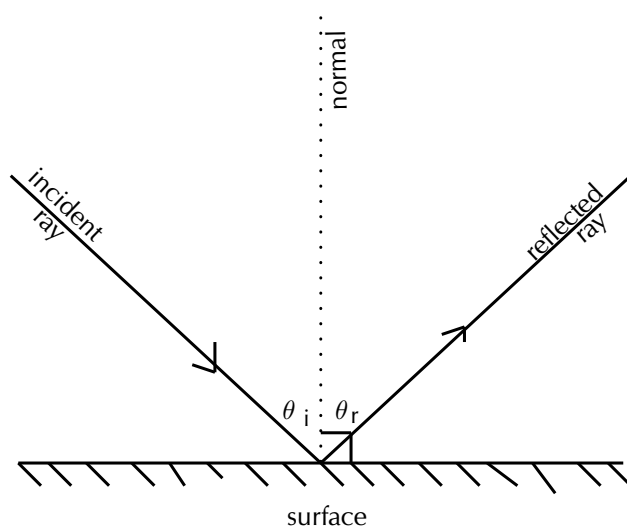


Figure 5.3: The angles of incidence and reflection are measured with respect to the normal to the surface.

When a ray of light is reflected, the reflected ray lies in the same plane as the incident ray and the normal. This plane is called the **plane of incidence** and is shown in Figure 5.4.

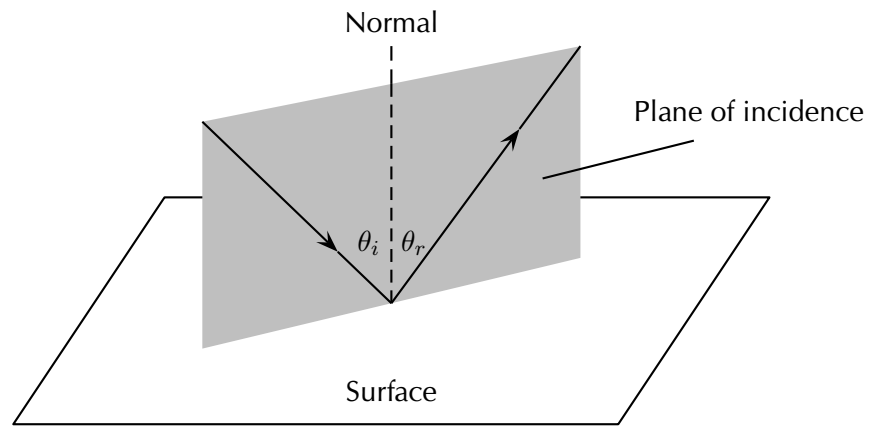


Figure 5.4: The plane of incidence is the plane including the incident ray and the normal to the surface. The reflected ray also lies in the plane of incidence.

DEFINITION: *Law of reflection*

The angle of incidence is equal to the angle of reflection:

$$\theta_i = \theta_r$$

and the incident ray, reflected ray, and the normal, all lie in the same plane.

The simplest example of the law of reflection is if the angle of incidence is 0° . In this case, the angle of reflection is also 0° . You see this when you look straight into a mirror.

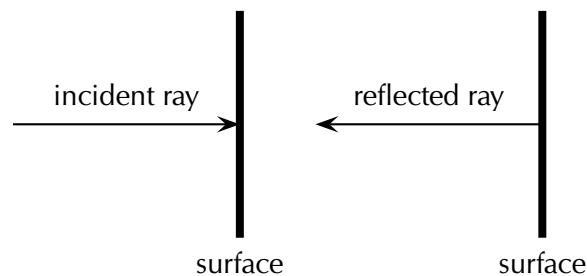


Figure 5.5: When a light ray strikes a surface at right angles to the surface, then the ray is reflected directly back.

Applying what we know from the law of reflection, if a light ray strikes a surface at 60° to the normal to the surface, then the angle that the reflected ray makes with the normal must also be 60° as shown in Figure 5.6.

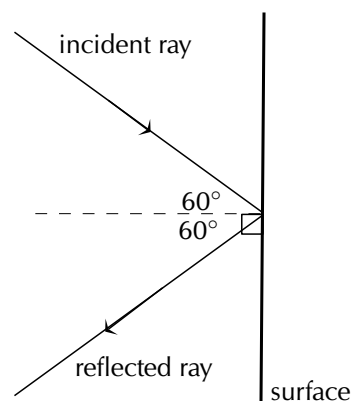


Figure 5.6: Ray diagram showing angle of incidence and angle of reflection. The law of reflection states that when a light ray reflects off a surface, the angle of reflection θ_r is the same as the angle of incidence θ_i .

Real world applications of reflection

A parabolic reflector is a mirror or dish (e.g. a satellite dish) which has a parabolic shape. Some examples of very useful parabolic reflectors are car headlamps, spotlights, telescopes and satellite dishes. In the case of car headlamps or spotlights, the outgoing light produced by the bulb is reflected by a parabolic mirror behind the bulb so that it leaves as a collimated beam (i.e. all the reflected rays are parallel). The reverse situation is true for a telescope where the incoming light from distant objects arrives as parallel rays and is focused by the parabolic mirror to a point, called the focus, where an image can be made. The surface of this sort of reflector has to be shaped very carefully so that the rays all arrive at the same focal point.

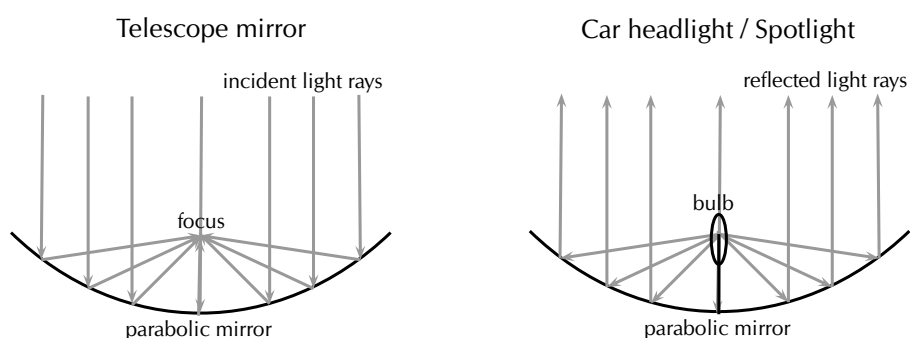
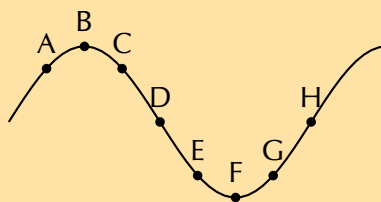


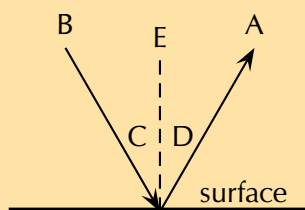
Figure 5.7: On the left is a ray diagram showing how a telescope mirror works to collect incoming incident light (parallel rays) from a distant object such as a star or galaxy and focus the rays to a point where a detector e.g. a camera, can make an image. The diagram on the right shows how the same kind of parabolic reflector can cause light coming from a car headlight or spotlight bulb to be collimated. In this case the reflected rays are parallel.

Exercise 5 – 1: Rays and Reflection

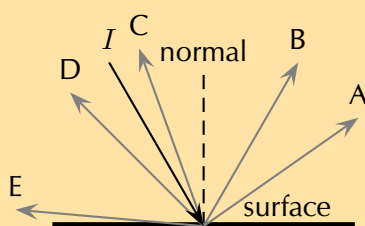
1. Are light rays real? Explain.
2. The diagram shows a curved surface. Draw normals to the surface at the marked points.



3. Which of the labels, A–H, in the diagram, correspond to the following:



- normal
 - angle of incidence
 - angle of reflection
 - incident ray
 - reflected ray
- State the Law of Reflection. Draw a diagram, label the appropriate angles and write a mathematical expression for the Law of Reflection.
 - Draw a ray diagram to show the relationship between the angle of incidence and the angle of reflection.
 - The diagram shows an incident ray *I*. Which of the other 5 rays (A, B, C, D, E) best represents the reflected ray of *I*?



- A ray of light strikes a surface at 15° to the normal to the surface. Draw a ray diagram showing the incident ray, reflected ray and surface normal. Calculate the angles of incidence and reflection and fill them in on your diagram.
- A ray of light leaves a surface at 45° to the normal to the surface. Draw a ray diagram showing the incident ray, reflected ray and surface normal. Calculate the angles of incidence and reflection and fill them in on your diagram.
- A ray of light strikes a surface at 25° to the surface. Draw a ray diagram showing the incident ray, reflected ray and surface normal. Calculate the angles of incidence and reflection and fill them in on your diagram.
- A ray of light leaves a surface at 65° to the surface. Draw a ray diagram showing the incident ray, reflected ray and surface normal. Calculate the angles of incidence and reflection and fill them in on your diagram.

11. A beam of light (for example from a torch) is generally not visible at night, as it travels through air. Try this for yourself. However, if you shine the torch through dust, the beam is visible. Explain why this happens.
12. If a torch beam is shone across a classroom, only students in the direct line of the beam would be able to see that the torch is shining. However, if the beam strikes a wall, the entire class will be able to see the spot made by the beam on the wall. Explain why this happens.
13. A scientist looking into a flat mirror hung perpendicular to the floor cannot see her feet but she can see the hem of her lab coat. Draw a ray diagram to help explain the answers to the following questions:
 - a) Will she be able to see her feet if she backs away from the mirror?
 - b) What if she moves towards the mirror?

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Absorption

In addition to being reflected, light can also be **absorbed**. Recall from grade 10 that visible light covers a range of wavelengths in the electromagnetic spectrum. The colours we see with our eyes correspond to light waves with different wavelengths or frequencies.

Imagine that you shine a torch or other source of white light onto a white piece of paper. The light is reflected off the paper and into our eyes and we see the colour of the paper as white. Now if we were to shine the same light onto a red apple we will notice that the colour of the apple appears red. This means that the surface of the apple is only reflecting red light into our eyes. All the other wavelengths in the incident white light are absorbed by the apple's skin. If you touch the apple, it will feel warm because it is absorbing the energy from all the light it is absorbing. Because of absorption and reflection, we can perceive colours of different objects. White objects reflect all or most of the wavelengths of light falling on them, coloured objects reflect particular wavelengths of light and absorb the rest. Black objects absorb all the light falling on them. This is why wearing a white t-shirt outside in the sun is cooler than wearing a black t-shirt, since the white t-shirt reflects most of the light falling on it, while the black t-shirt will absorb it and heat up.

Transmission

A further property of light is that it can be **transmitted** through objects or a medium. Objects through which light can be transmitted are called *transparent* and objects

which block out light or that light cannot pass through are called *opaque*.

For example, glass windows allow visible light to pass through them which is why we can see through windows. The light rays from things outside the window can pass through or be transmitted through the glass and into our eyes. Brick walls on the other hand are opaque to visible light. We cannot see through brick walls because the light cannot be transmitted through the wall into our eyes. The transmission of light through an object depends on the wavelength of the light. For example, short wavelength visible light cannot be transmitted through a brick wall whereas long wavelength radio waves can easily pass through walls and be received by a radio or cell phone. In other words, a brick wall is transparent to radio waves!

5.4 The speed of light

ESBN3

One of the most exciting discoveries in physics during the last century, and the cornerstone of Einstein's Theory of Relativity, is that light travels at a constant speed in a given medium. Light also has a maximum speed at which it can propagate, and nothing can move faster than the speed of light. The maximum speed at which light can travel is when it propagates through free space (a vacuum) at $299\,792\,485\text{ m}\cdot\text{s}^{-1}$. A vacuum is a region with no matter in it, not even air. However, the speed of light in air is very close to that in a vacuum.

We use the symbol c to represent the speed of light in a vacuum and approximate it as $c = 3 \times 10^8\text{ m}\cdot\text{s}^{-1}$

DEFINITION: *Speed of light*

The speed of light, c , is constant in a given medium and has a maximum speed in vacuum of

$$c = 3 \times 10^8\text{ m}\cdot\text{s}^{-1}$$

5.5 Refraction

ESBN4

▶ See video: 23SD at www.everythingscience.co.za

▶ See video: 23SF at www.everythingscience.co.za

In the previous sections we studied light reflecting off various surfaces. What happens when light passes from one medium into another? The speed of light, like that of all waves, is dependent on the medium through which it is travelling. When light moves from one medium into another (for example, from air to glass), the speed of light changes. If the light ray hits the boundary of the new medium (for example the edge of a glass block) at any angle which is not perpendicular to or parallel with the boundary, the light ray will change its direction through the next medium, or appear to 'bend'. This is called **refraction** of light. It is important to note that while the speed of the light changes when it passes into the new medium, the frequency of the light

remains the same.

DEFINITION: *Refraction*

Refraction occurs at the boundary of two media when light travels from one medium into the other and its speed changes but its frequency remains the same. If the light ray hits the boundary at an angle which is not perpendicular to or parallel to the surface, then it will change direction and appear to 'bend'.

Refraction is nicely demonstrated when you look from above at an angle at a straw in a glass of water. The straw appears bent in the liquid. This is because the light rays leaving the straw change direction when they hit the surface between the liquid and the air. Your eyes trace the light rays backwards as straight lines to the point they would have come from if they had not changed direction and as a result you see the tip of the straw as being shallower in the liquid than it really is.

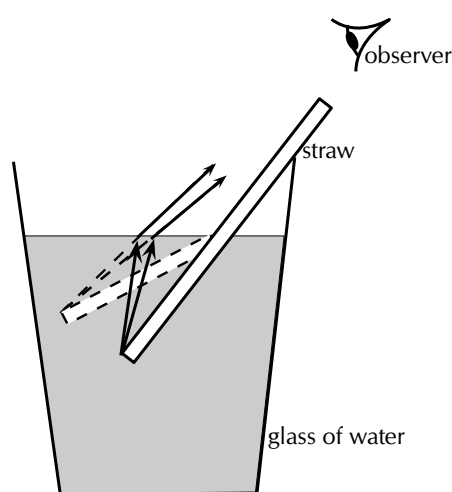


Figure 5.8: Due to refraction, a straw in a glass of water appears bent when an observer looks down at an angle from above the water surface.

Refractive index

ESBN5

The speed of light and therefore the degree of bending of the light depends on the *refractive index* of material through which the light passes. The refractive index (symbol n) is the ratio of the speed of light in a vacuum to its speed in the material.

DEFINITION: *Refractive index*

The refractive index (symbol n) of a material is the ratio of the speed of light in a vacuum to its speed in the material and gives an indication of how difficult it is for light to get through the material.

$$n = \frac{c}{v}$$

where

n = refractive index (no unit)

c = speed of light in a vacuum ($3,00 \times 10^8 \text{ m}\cdot\text{s}^{-1}$)

v = speed of light in a given medium ($\text{m}\cdot\text{s}^{-1}$)

Using the definition of refractive index

$$n = \frac{c}{v}$$

we can see how the speed of light changes in different media, because the speed of light in a vacuum, c , is constant.

If the refractive index, n , increases, the speed of light in the material, v , must decrease. Therefore light travels slower through materials of high refractive index, n .

Table 5.1 shows refractive indices for various materials. Light travels slower in any material than it does in a vacuum, so all values for n are greater than 1.

Medium	Refractive Index
Vacuum	1
Helium	1,000036
Air*	1,0002926
Carbon dioxide	1,00045
Water: Ice	1,31
Water: Liquid (20°C)	1,333
Acetone	1,36
Ethyl Alcohol (Ethanol)	1,36
Sugar solution (30%)	1,38
Fused quartz	1,46
Glycerine	1,4729
Sugar solution (80%)	1,49
Rock salt	1,516
Crown Glass	1,52
Sodium chloride	1,54
Polystyrene	1,55 to 1,59
Bromine	1,661
Sapphire	1,77
Glass (typical)	1,5 to 1,9
Cubic zirconia	2,15 to 2,18
Diamond	2,419
Silicon	4,01

Table 5.1: Refractive indices of some materials. n_{air} is calculated at standard temperature and pressure (STP).

Worked example 1: Refractive index

QUESTION

Calculate the speed of light through glycerine which has a refractive index of 1,4729.

SOLUTION

Step 1: Determine what is given and what is required

We are given the refractive index, n of glycerine and we need to calculate the speed of light in glycerine.

Step 2: Determine how to approach the problem

We can use the definition of refractive index since the speed of light in vacuum is a constant and we know the value of glycerine's refractive index.

Step 3: Do the calculation

$$n = \frac{c}{v}$$

Rearrange the equation to solve for v and substitute in the known values:

$$\begin{aligned} v &= \frac{c}{n} \\ &= \frac{3 \times 10^8 \text{ m}\cdot\text{s}^{-1}}{1,4729} \\ &= 2,04 \times 10^8 \text{ m}\cdot\text{s}^{-1} \end{aligned}$$

Exercise 5 – 2: Refractive index

1. Use the values given in Table 5.1, and the definition of refractive index to calculate the speed of light in water (ice).
2. Calculate the refractive index of an unknown substance where the speed of light through the substance is $1,974 \times 10^8 \text{ m}\cdot\text{s}^{-1}$. Round off your answer to 2 decimal places. Using Table 5.1, identify what the unknown substance is.

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Optical density

ESBN6

Optical density is a measure of the refracting power of a medium. In other words, the higher the optical density, the more the light will be refracted or slowed down as it moves through the medium. Optical density is related to refractive index in that materials with a high refractive index will also have a high optical density. Light will travel slower through a medium with a high refractive index and high optical density and faster through a medium which has a low refractive index and a low

optical density.

DEFINITION: *Optical density*

Optical density is a measure of the refracting power of a medium.

Representing refraction with ray diagrams

ESBN7

It is useful to draw ray diagrams to understand how the geometrical optics concepts we have discussed previously work. Before we can draw the diagrams we need to define a few concepts such as the normal to a surface, the angle of incidence and the angle of refraction.

DEFINITION: *Normal*

The normal to a surface is the line which is perpendicular to the plane of the surface.

DEFINITION: *Angle of incidence*

The angle of incidence is the angle defined between the normal to a surface and the incoming (incident) light ray.

DEFINITION: *Angle of refraction*

The angle of refraction is the angle defined between the normal to a surface and the refracted light ray.

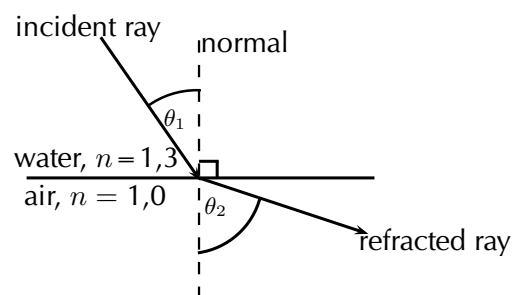


Figure 5.9: The diagram shows the boundary between two media: water and air. An incoming light ray is refracted when it passes through the surface of the water into the air. The angle of incidence is θ_1 and the angle of refraction is θ_2 .

When light travels from one medium to another, it is refracted. If the angle of incidence is not equal to zero, the light ray will change direction from its original path as it is refracted.

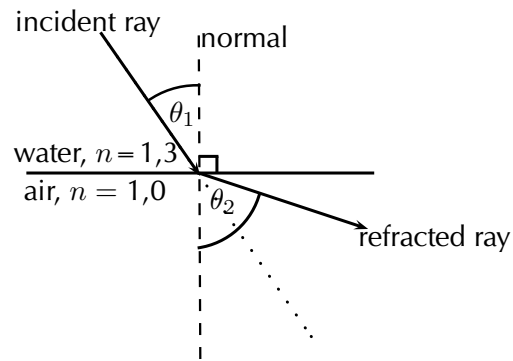


Figure 5.10: Light is moving from an optically dense medium to an optically less dense medium.

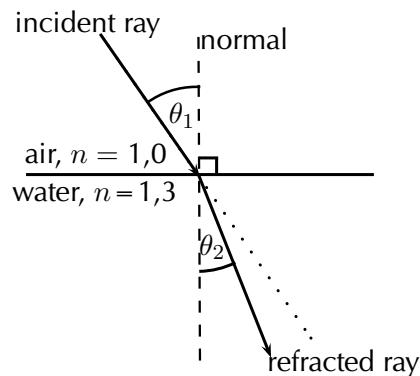


Figure 5.11: Light is moving from an optically less dense medium to an optically denser medium.

A practical demonstration of the propagation of light from air into glass and back into air or light from one medium to another can be done. You will need a glass block, ray box, colour filters, non-rectangular glass blocks, water, paper, ruler, pencil and protractor.

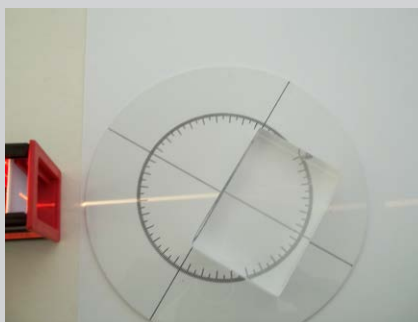
General experiment: Propagation of light from air into glass and back into air

Aim:

To investigate propagation of light from air into glass and back into air

Apparatus:

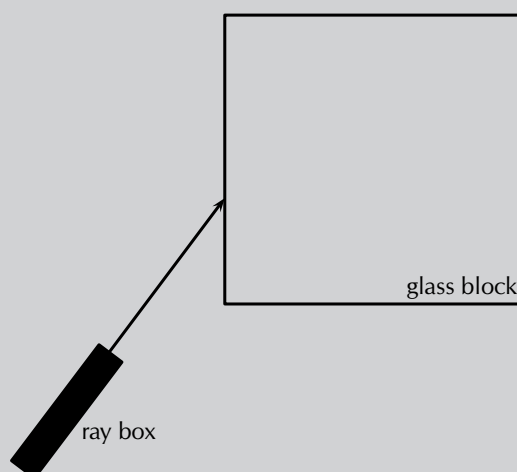
ray box, rectangular glass block, plain paper, pencil, ruler, protractor.



Method:

Follow the steps below:

1. Place the glass block on a plain piece of paper and use your pencil to draw around the block to make its outline on the paper.
2. Now turn on the ray box and aim the light ray through the left side of the glass block as illustrated in the diagram:



3. Now use your pencil to draw a dot somewhere on the incident light ray and another dot at the point where it enters the glass block.
4. Use your pencil to draw a dot at the point where the light exits the glass block and also somewhere else along the exiting ray.
5. Turn off the ray box and remove the glass block from the paper. Use your ruler to join the dots so that you have drawn a picture that looks like the figure above.
6. Now draw the normals to the surfaces where the light ray enters and leaves the block and mark the angle of incidence and angle of refraction on the left surface and the right surface.

Questions for discussion:

1. At the surface where the light enters the glass block, what do you notice about the angle of incidence compared to the angle of refraction?

2. Now look at the surface where the light exits the glass block. Compare the angle of incidence and angle of refraction here.
3. How do the optical densities and indices of refraction for air and glass compare?

Informal experiment: Propagation of light from one medium into another medium

Aim:

To investigate the propagation of light from one medium into another

Apparatus:

ray box, glass blocks of various shapes, a transparent container filled with water, plain paper, pencil, ruler, protractor.

Method:

Starting with the rectangular glass block, repeat the steps below for each of the various differently shaped glass blocks:

1. Place the glass block on a plain piece of paper and use your pencil to draw around the block to make its outline on the paper.
2. Turn on the ray box and aim the light ray through one of the block's surfaces.
3. Draw a dot at the point where the light enters the block and another dot somewhere else along the incident ray. Also draw a dot at the point where the ray exits the surface of the block and another somewhere along the exiting ray.
4. Remove the glass block and turn off the ray box. Use your ruler to join the dots.
5. Now draw the normals to the surfaces where the light ray enters and exits the block.
6. Use your protractor to measure the angles of incidence and refraction at the surfaces where the light ray enters and exits the block.

Now follow the same steps as before but place differently coloured filters at the surface of the blocks where the light enters the block.

Lastly, replace the glass blocks with the container of water and repeat the same steps as above. For this case, try to aim the ray box so that the angle of incidence on the water container is the same as it was for the rectangular glass block you investigated first. This is easier if you place your new piece of paper on top of the drawing you made with the rectangular block. Line up the water container where you drew the edge of your glass block. Then you can aim the ray box so that the light lines up with the incident ray on your paper and follow the steps as before.

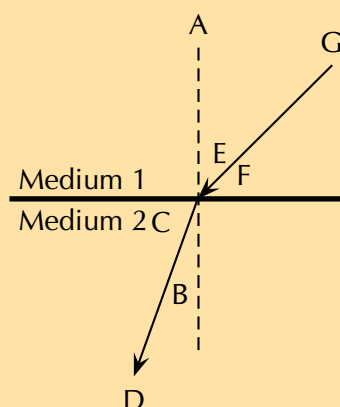
Questions for discussion:

1. For each of your differently shaped glass blocks, how does the angle of incidence going from air to glass compare to the angle of refraction?
2. What happens when you place a coloured filter at the boundary between the air and a glass block?
3. When you compare your two diagrams for the rectangular glass block and the water container, if you did it correctly, the angles of incidence from air to glass/water should be the same. What can you say about the angles of refraction for glass and water respectively? How does this compare with what you already know about the refractive indices or optical densities of these materials?

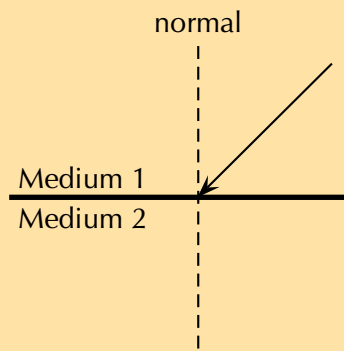
🎧 See simulation: 23SJ at www.everythingscience.co.za

Exercise 5 – 3: Refraction

1. Explain refraction in terms of a change of wave speed in different media.
2. In the diagram, label the following:
 - a) angle of incidence
 - b) angle of refraction
 - c) incident ray
 - d) refracted ray
 - e) normal



3. What is *the angle of refraction*?
4. Describe what is meant by *the refractive index of a medium*.
5. In the diagram, a ray of light strikes the interface between two media.



Draw what the refracted ray would look like if:

- a) medium 1 had a higher refractive index than medium 2.
 - b) medium 1 had a lower refractive index than medium 2.
6. **Challenge question:** What values of n are physically impossible to achieve? Explain your answer. The values provide the limits of possible refractive indices.
7. **Challenge question:** You have been given a glass beaker full of an unknown liquid. How would you identify what the liquid is? You have the following pieces of equipment available for the experiment: a laser or ray box, a protractor, a ruler, a pencil, and a reference guide containing optical properties of various liquids.

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5.6 Snell's Law

ESBN8

Now that we know that the degree of bending, or the angle of refraction, is dependent on the refractive index of a medium, how do we calculate the angle of refraction? The answer to this question was discovered by a Dutch physicist called Willebrord Snell in 1621 and is now called Snell's Law or the Law of Refraction.

DEFINITION: *Snell's law*

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

where

n_1 = Refractive index of material 1

n_2 = Refractive index of material 2

θ_1 = Angle of incidence

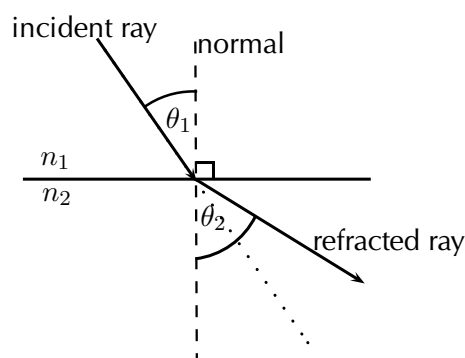
θ_2 = Angle of refraction

FACT

Snell never published his discovery of the Law of Refraction. His work was actually published by another prominent physicist of the time, Christiaan Huygens, who gave credit to Snell.

Remember that angles of incidence and refraction are measured from the normal, which is an imaginary line perpendicular to the surface.

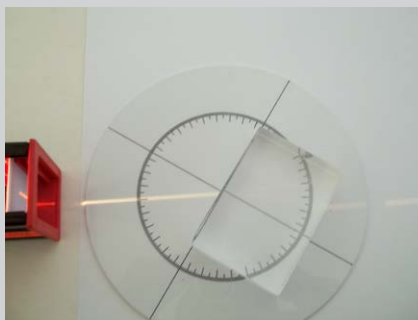
Suppose we have two media with refractive indices n_1 and n_2 . A light ray is incident on the surface between these materials with an **angle of incidence** θ_1 . The refracted ray that passes through the second medium will have an **angle of refraction** θ_2 .

**Formal experiment: Verifying Snell's Law****Aim:**

To verify Snell's law

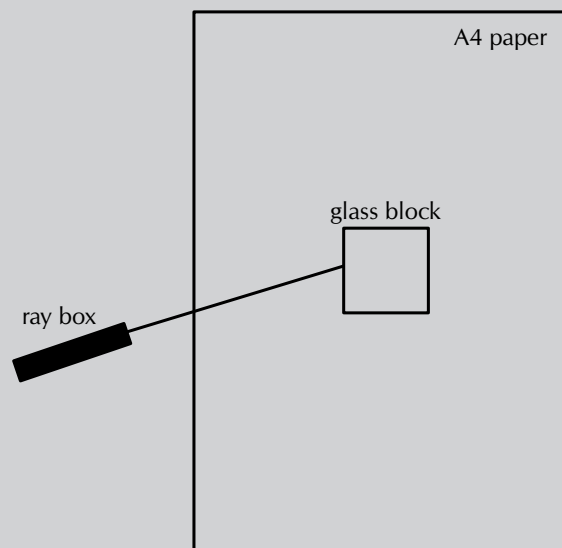
Apparatus:

glass block, ray box, 360° protractor, 5 pieces of A4 paper, pencil, ruler

**Method:**

This experiment will require you to follow the steps below 5 times (once for each piece of A4 paper).

1. Place the glass block in the middle of the A4 piece of paper so that its sides are parallel to each of the sides of the paper and draw around the block with a pencil to make its outline on the piece of paper.
2. Turn on the ray box and aim the light ray towards the glass block so that it makes an angle with the nearest surface of the block as shown in the picture. For each piece of paper, change the angle of the incoming ray.

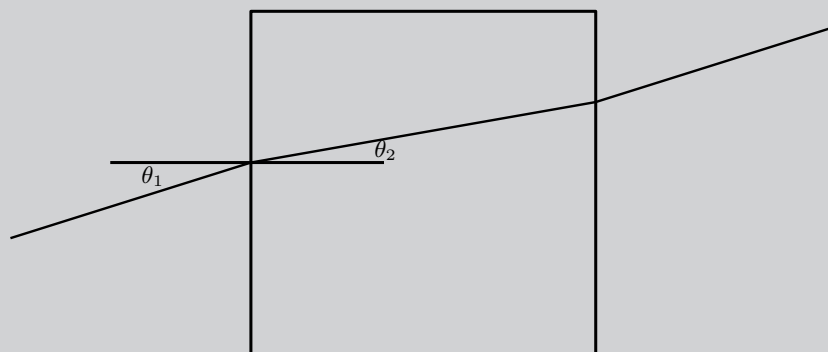


3. You will now need to mark on the paper, the path of the incoming and outgoing light rays. Do this by first drawing a dot on the paper somewhere along the incoming light ray. Now draw a second dot on the paper at the point where the incoming light ray hits the surface of the block. Do the same thing for the outgoing light ray; mark the point where it leaves the block and some other point along its path.
4. Now switch off the ray box and remove the glass block from the paper. Use a ruler to join the dots of the incoming ray. Now join the dots of the outgoing ray. Lastly, draw a line which joins the point where the incoming ray hits the block and where the outgoing ray leaves the block. This is the path of the light ray through the glass.
5. The aim of this experiment is to verify Snell's law. i.e. $n_1 \sin \theta_1 = n_2 \sin \theta_2$. We know the refractive index of our two media:

For air, $n_1 = 1,0$

For glass, $n_2 = 1,5$

Now we need to measure the two angles, θ_1 and θ_2 . To do this, we need to draw the normal to the surface where the light ray enters the block. Use the protractor to measure an angle of 90° to the entry surface and draw the normal. At this point, the drawing on your piece of paper should look something like the picture:



6. Now measure θ_1 and θ_2 using the protractor. Enter the values you measured into a table which looks like:

θ_1	θ_2	$n_1 \sin \theta_1$	$n_2 \sin \theta_2$

Discussion:

Have a look at your completed table. You should have 5 rows filled in, one for each of your pieces of A4 paper. For each row, what do you notice about the values in the last two columns? Do your values agree with what Snell's law predicts?

Formal experiment: Using Snell's law to determine the refractive index of an unknown material

Aim:

To determine the refractive index of an unknown material

Apparatus:

ray box, 360° protractor, 5 pieces of A4 paper, a block of unknown transparent material, pencil, ruler

Method:

This experiment will require you to follow the steps below at least 5 times. The steps are the same as you followed in the previous experiment.

1. Place the block in the middle of the A4 piece of paper so that its sides are parallel to each of the sides of the paper and draw around the block with a pencil to make its outline on the piece of paper.
2. Turn on the ray box and aim the light ray towards the block so that it makes an angle with the nearest surface of the block. For each piece of paper, change the angle of the incoming ray.
3. You will now need to mark on the paper, the path of the incoming and outgoing light rays. Do this by first drawing a dot on the paper somewhere along the incoming light ray. Now draw a second dot on the paper at the point where the incoming light ray hits the surface of the block. Do the same thing for the outgoing light ray; mark the point where it leaves the block and some other point along its path.
4. Now switch off the ray box and remove the block from the paper. Use a ruler to join the dots of the incoming ray. Now join the dots of the outgoing ray. Lastly, draw a line which joins the point where the incoming ray hits the block and where the outgoing ray leaves the block. This is the path of the light ray through the block.

5. The aim of this experiment is to determine the refractive index n_2 of the unknown material using Snell's law which states $n_1 \sin \theta_1 = n_2 \sin \theta_2$. We know that for air $n_1 = 1,0$ and we can measure the angle of incidence, θ_1 and the angle of refraction, θ_2 . Then we can solve the equation for the unknown, n_2 .

To measure θ_1 and θ_2 , we need to draw the normal to the surface where the light ray enters the block. Use the protractor to measure an angle of 90° to the entry surface and draw the normal. Now use the protractor to measure θ_1 and θ_2 .

6. Enter the values you measured into a table which looks like the one below and calculate the resulting value for n_2 .

θ_1	θ_2	$n_2 = \frac{n_1 \sin \theta_1}{\sin \theta_2}$

Question:

What do you notice about all your values in the last column of the table?

Discussion:

You should have noticed that the values in the last column of the table are similar but not identical. This is due to measurement errors when you measured the angles of incidence and angles of refraction. These sorts of errors are common in all physics experiments and lead to a measure of uncertainty in the final extracted value. However, since we did the same experiment 5 times, we can average the 5 independent measurements of n_2 to get a good approximation to the real value for our unknown material.

If

$$n_2 > n_1$$

then from Snell's Law,

$$\sin \theta_1 > \sin \theta_2$$

For angles smaller than 90° , $\sin \theta$ increases as θ increases. Therefore,

$$\theta_1 > \theta_2.$$

This means that the angle of incidence is greater than the angle of refraction and the light ray is bent toward the normal.

Similarly, if

$$n_2 < n_1$$

then from Snell's Law,

$$\sin \theta_1 < \sin \theta_2.$$

For angles smaller than 90° , $\sin \theta$ increases as θ increases. Therefore,

$$\theta_1 < \theta_2.$$

This means that the angle of incidence is less than the angle of refraction and the light ray is bent away from the normal.

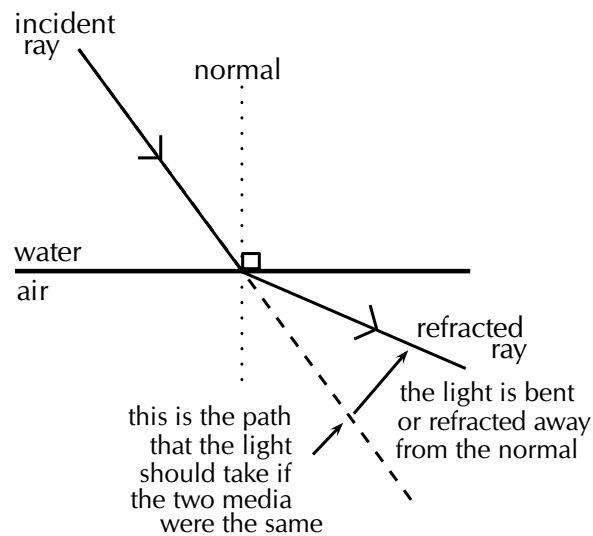


Figure 5.12: Light is moving from a medium with a higher refractive index to one with a lower refractive index. Light is refracted away from the normal.

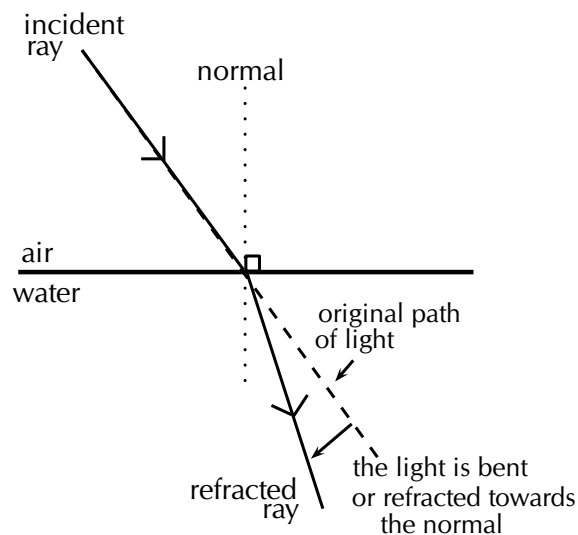


Figure 5.13: Light is moving from a medium with a lower refractive index to a medium with a higher refractive index. Light is refracted towards the normal.

What happens to a ray that lies along the normal line? In this case, the angle of incidence is 0° and

$$\begin{aligned}\sin \theta_2 &= \frac{n_1}{n_2} \sin \theta_1 \\ &= 0 \\ \therefore \theta_2 &= 0.\end{aligned}$$

This shows that if the light ray is incident at 0° , then the angle of refraction is also 0° . The direction of the light ray is unchanged, however, the speed of the light will change as it moves into the new medium. Therefore refraction still occurs although it will not be easily observed.

Let's use an everyday example that you can more easily imagine. Imagine you are pushing a lawnmower or a cart through short grass. As long as the grass is pretty much the same length everywhere, it is easy to keep the mower or cart going in a straight line. However, if the wheels on one side of the mower or cart enter an area of grass that is longer than the grass on the other side, that side of the mower will move more slowly since it is harder to push the wheels through the longer grass. The result is that the mower will start to turn inwards, towards the longer grass side as you can see in the picture. This is similar to light which passes through a medium and then enters a new medium with a higher refractive index or higher optical density. The light will change direction towards the normal, just like the mower in the longer grass. The opposite happens when the mower moves from an area of longer grass into an area with shorter grass. The side of the mower in the shorter grass will move faster and the mower will turn outwards, just like a light ray moving from a medium with high refractive index into a medium with low refractive index moves away from the normal.

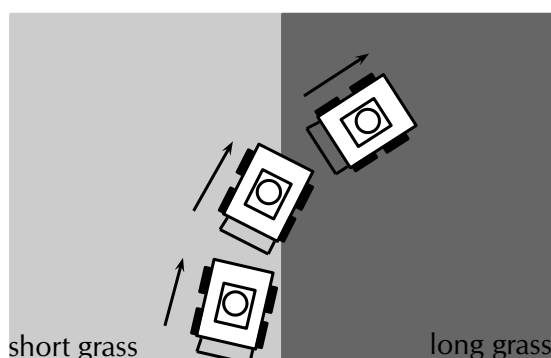


Figure 5.14: The lawnmower is moving from an area of short grass into an area of longer grass. When it reaches the boundary between the areas, the wheels which are in the long grass move slower than the wheels which are still in the short grass, causing the lawnmower to change direction and its path to bend inwards.

Worked example 2: Using Snell's Law

QUESTION

Light is refracted at the boundary between water and an unknown medium. If the angle of incidence is 25° , and the angle of refraction is $20,6^\circ$, calculate the refractive index of the unknown medium and use Table 5.1 to identify the material.

SOLUTION

Step 1: Determine what is given and what is being asked

The angle of incidence $\theta_1 = 25^\circ$

The angle of refraction $\theta_2 = 20,6^\circ$

We can look up the refractive index for water in Table 5.1: $n_1 = 1,333$

We need to calculate the refractive index for the unknown medium and identify it.

Step 2: Determine how to approach the problem

We can use Snell's law to calculate the unknown refractive index, n_2

According to Snell's Law:

$$\begin{aligned}n_1 \sin \theta_1 &= n_2 \sin \theta_2 \\n_2 &= \frac{n_1 \sin \theta_1}{\sin \theta_2} \\n_2 &= \frac{1,333 \sin 25^\circ}{\sin 20,6^\circ} \\n_2 &= 1,6\end{aligned}$$

Step 3: Identify the unknown medium

According to Table 5.1, typical glass has a refractive index between 1,5 to 1,9. Therefore the unknown medium is typical glass.

Worked example 3: Using Snell's law

QUESTION

A light ray with an angle of incidence of 35° passes from water to air. Find the angle of refraction using Snell's Law and Table 5.1. Discuss the meaning of your answer.

SOLUTION

Step 1: Determine the refractive indices of water and air

From Table 5.1, the refractive index is 1,333 for water and about 1 for air. We know the angle of incidence, so we are ready to use Snell's Law.

Step 2: Substitute values

According to Snell's Law:

$$\begin{aligned}n_1 \sin \theta_1 &= n_2 \sin \theta_2 \\1,33 \sin 35^\circ &= 1 \sin \theta_2 \\\sin \theta_2 &= 0,763 \\\theta_2 &= 49,7^\circ \text{ or } 130,3^\circ\end{aligned}$$

Since $130,3^\circ$ is larger than 90° , the solution is:

$$\theta_2 = 49,7^\circ$$

Step 3: Discuss the answer

The light ray passes from a medium of high refractive index to one of low refractive index. Therefore, the light ray is bent away from the normal.

Worked example 4: Using Snell's Law

QUESTION

A light ray passes from water to diamond with an angle of incidence of 75° . Calculate the angle of refraction. Discuss the meaning of your answer.

SOLUTION

Step 1: Determine the refractive indices of water and air

From Table 5.1, the refractive index is 1,333 for water and 2,42 for diamond. We know the angle of incidence, so we are ready to use Snell's Law.

Step 2: Substitute values and solve

According to Snell's Law:

$$\begin{aligned}n_1 \sin \theta_1 &= n_2 \sin \theta_2 \\1,33 \sin 75^\circ &= 2,42 \sin \theta_2 \\ \sin \theta_2 &= 0,531 \\ \theta_2 &= 32,1^\circ\end{aligned}$$

Step 3: Discuss the answer

The light ray passes from a medium of low refractive index to one of high refractive index. Therefore, the light ray is bent towards the normal.

Exercise 5 – 4: Snell's Law

1. State Snell's Law.
2. Light travels from a region of glass into a region of glycerine, making an angle of incidence of 40° .



- a) Draw the incident and refracted light rays on the diagram and label the angles of incidence and refraction.
- b) Calculate the angle of refraction.

3. A ray of light travels from silicon to water. If the ray of light in the water makes an angle of 69° to the normal to the surface, what is the angle of incidence in the silicon?
4. Light travels from a medium with $n = 1,25$ into a medium of $n = 1,34$, at an angle of 27° from the normal.
 - a) What happens to the speed of the light? Does it increase, decrease, or remain the same?
 - b) What happens to the wavelength of the light? Does it increase, decrease, or remain the same?
 - c) Does the light bend towards the normal, away from the normal, or not at all?
5. Light travels from a medium with $n = 1,63$ into a medium of $n = 1,42$.
 - a) What happens to the speed of the light? Does it increase, decrease, or remain the same?
 - b) What happens to the wavelength of the light? Does it increase, decrease, or remain the same?
 - c) Does the light bend towards the normal, away from the normal, or not at all?
6. Light is incident on a rectangular glass prism. The prism is surrounded by air. The angle of incidence is 23° . Calculate the angle of reflection and the angle of refraction.
7. Light is refracted at the interface between air and an unknown medium. If the angle of incidence is 53° and the angle of refraction is 37° , calculate the refractive index of the unknown, second medium.
8. Light is refracted at the interface between a medium of refractive index 1,5 and a second medium of refractive index 2,1. If the angle of incidence is 45° , calculate the angle of refraction.
9. A ray of light strikes the interface between air and diamond. If the incident ray makes an angle of 30° with the interface, calculate the angle made by the refracted ray with the interface.
10. The angles of incidence and refraction were measured in five unknown media and recorded in the table below. Use your knowledge about Snell's Law to identify each of the unknown media A–E. Use Table 5.1 to help you.

Medium 1	n_1	θ_1	θ_2	n_2	Unknown Medium
Air	1,0002926	38	27	?	A
Air	1,0002926	65	38,4	?	B
Vacuum	1	44	16,7	?	C
Air	1,0002926	15	6,9	?	D
Vacuum	1	20	13,3	?	E

11. Zingi and Tumi performed an investigation to identify an unknown liquid. They shone a ray of light into the unknown liquid, varying the angle of incidence and recording the angle of refraction. Their results are recorded in the following table:

Angle of incidence	Angle of refraction
0,0°	0,00°
5,0°	3,76°
10,0°	7,50°
15,0°	11,2°
20,0°	14,9°
25,0°	18,5°
30,0°	22,1°
35,0°	25,5°
40,0°	28,9°
45,0°	32,1°
50,0°	35,2°
55,0°	38,0°
60,0°	40,6°
65,0°	43,0°
70,0°	?
75,0°	?
80,0°	?
85,0°	?

- Write down an aim for the investigation.
- Make a list of all the apparatus they used.
- Identify the unknown liquid.

12. Predict what the angle of refraction will be for 70°, 75°, 80° and 85°.

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. [23ST](#) 2. [23SV](#) 3. [23SW](#) 4. [23SX](#) 5. [23SY](#) 6. [23SZ](#)
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5.7 Critical angles and total internal reflection

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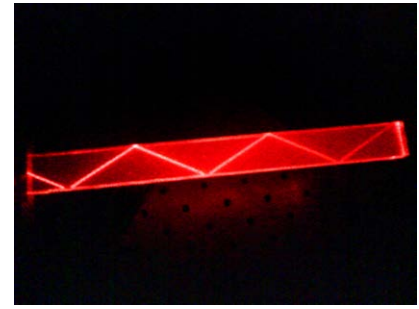
Total internal reflection

ESBNB

▶ See video: [23T8](#) at www.everythingscience.co.za

You may have noticed when experimenting with ray boxes and glass blocks in the previous section that sometimes, when you changed the angle of incidence of the light, it was not refracted out into the air, but was reflected back through the block.

When the entire incident light ray travelling through an optically denser medium is reflected back at the boundary between that medium and another of lower optical density, instead of passing through and being refracted, this is called total internal reflection.



As we increase the angle of incidence, we reach a point where the angle of refraction is 90° and the refracted ray travels along the boundary of the two media. This angle of incidence is called the critical angle.

DEFINITION: *Critical angle*

The critical angle is the angle of incidence where the angle of refraction is 90° . The light must travel from an optically more dense medium to an optically less dense medium.

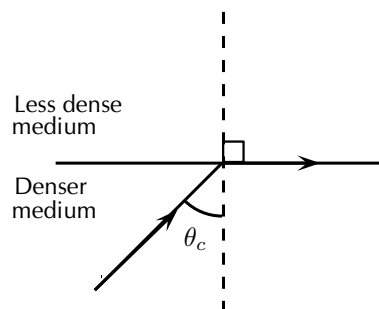


Figure 5.15: When the angle of incidence is equal to the critical angle, the angle of refraction is equal to 90° .

If the angle of incidence is bigger than this critical angle, the refracted ray will not emerge from the medium, but will be reflected back into the medium. This is called **total internal reflection**.

The conditions for total internal reflection are:

1. light is travelling from an optically denser medium (higher refractive index) to an optically less dense medium (lower refractive index).
2. the angle of incidence is greater than the critical angle.

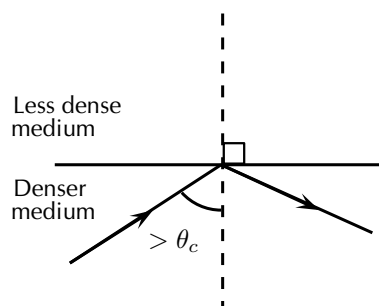


Figure 5.16: When the angle of incidence is greater than the critical angle, the light ray is reflected at the boundary of the two media and total internal reflection occurs.

Each pair of media have their own unique critical angle. For example, the critical angle for light moving from glass to air is 42° , and that of water to air is $48,8^\circ$.

Informal experiment: The critical angle of a rectangular glass block

Aim:

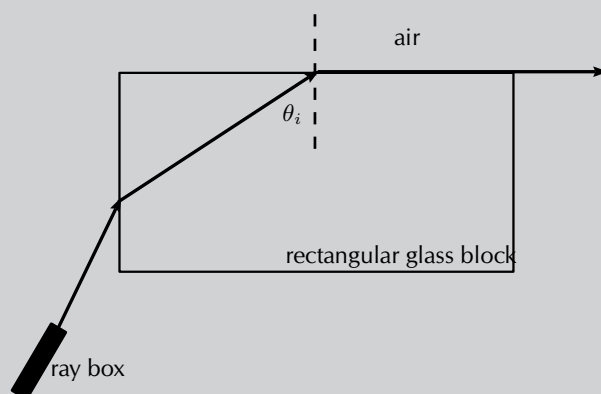
To determine the critical angle for a rectangular glass block.

Apparatus:

rectangular glass block, ray box, 360° protractor, paper, pencil, ruler

Method:

1. Place the glass block in the middle of the piece of paper and draw around the outside of the block with your pencil to make its outline.
2. Turn on the ray box and aim the light ray into the left side of the glass block. Adjust the angle at which the light strikes the glass block until you see the refracted light ray travelling along the top edge of the glass block (i.e. the angle of refraction is 90°). This situation should look something like the following diagram:



3. Draw a dot on the paper at the point where the light enters the glass block from the ray box. Then draw a dot on the paper at the point where the light is refracted (at the top of the glass block).
4. Turn off the ray box and remove the glass block from the paper.
5. Now use your ruler to draw a line between the two dots. This line represents the incident light ray.
6. When the angle of refraction is 90° , the angle of incidence is equal to the critical angle. Therefore to determine the critical angle, we need to measure this angle of incidence. Do this using your protractor.

TIP

Remember that for total internal reflection the incident ray is always in the denser medium!

7. Compare your answer with the values other members of your class obtain and discuss why they might not be identical (although they should be similar!).

Calculating the critical angle

Instead of always having to measure the critical angles of different materials, it is possible to calculate the critical angle at the surface between two media using Snell's Law. To recap, Snell's Law states:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

where n_1 is the refractive index of material 1, n_2 is the refractive index of material 2, θ_1 is the angle of incidence and θ_2 is the angle of refraction. For total internal reflection we know that the angle of incidence is the critical angle. So,

$$\theta_1 = \theta_c.$$

However, we also know that the angle of refraction at the critical angle is 90° . So we have:

$$\theta_2 = 90^\circ.$$

We can then write Snell's Law as:

$$n_1 \sin \theta_c = n_2 \sin 90^\circ$$

Solving for θ_c gives:

$$n_1 \sin \theta_c = n_2 \sin 90^\circ$$

$$\sin \theta_c = \frac{n_2}{n_1}(1)$$

$$\therefore \theta_c = \sin^{-1} \left(\frac{n_2}{n_1} \right)$$

Worked example 5: Critical angle

QUESTION

Given that the refractive indices of air and water are 1,00 and 1,33 respectively, find the critical angle.

SOLUTION

Step 1: Determine how to approach the problem

We can use Snell's law to determine the critical angle since we know that when the angle of incidence equals the critical angle, the angle of refraction is 90° .

Step 2: Solve the problem

$$\begin{aligned}n_1 \sin \theta_c &= n_2 \sin 90^\circ \\ \theta_c &= \sin^{-1} \left(\frac{n_2}{n_1} \right) \\ &= \sin^{-1} \left(\frac{1}{1,33} \right) \\ &= 48,8^\circ\end{aligned}$$

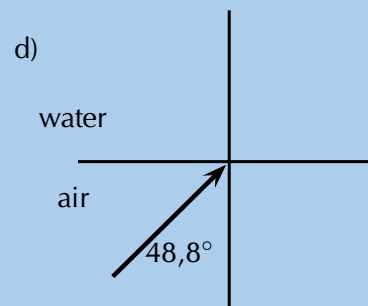
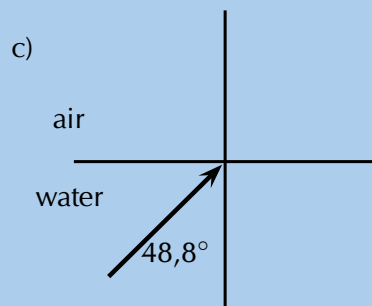
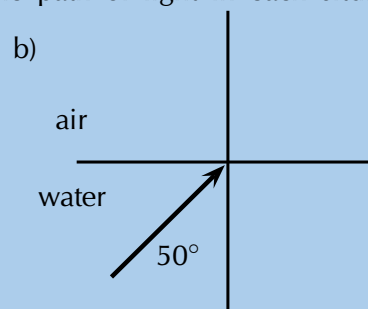
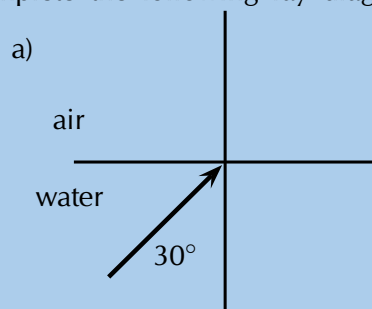
Step 3: Write the final answer

The critical angle for light travelling from water to air is $48,8^\circ$.

Worked example 6: Critical angle

QUESTION

Complete the following ray diagrams to show the path of light in each situation.



SOLUTION

Step 1: Identify what is given and what is asked

The critical angle for water is $48,8^\circ$

We are asked to complete the diagrams.

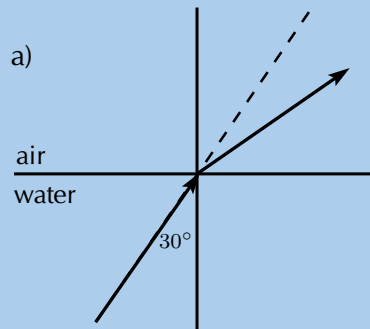
For incident angles smaller than $48,8^\circ$ refraction will occur.

For incident angles greater than $48,8^\circ$ total internal reflection will occur.

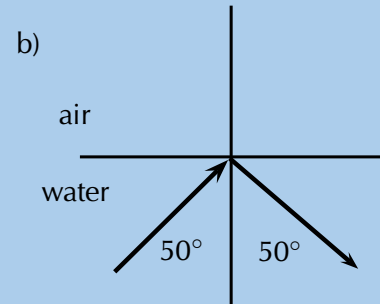
For incident angles equal to $48,8^\circ$ refraction will occur at 90° .

The light must travel from a medium with a higher refractive index (higher optical density) to a medium with lower refractive index (lower optical density).

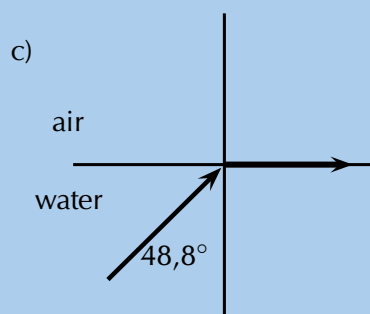
Step 2: Complete the diagrams



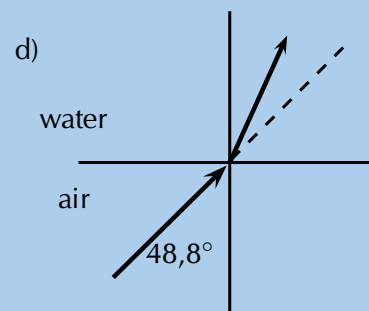
Refraction occurs (ray is bent away from the normal).



Total internal reflection occurs.



$\theta_c = 48,8^\circ$.



Refraction towards the normal (air is less dense than water).

Fibre optics

Total internal reflection is a very useful natural phenomenon since it can be used to confine light. One of the most common applications of total internal reflection is in *fibre optics*. An optical fibre is a thin, transparent fibre, usually made of glass or plastic, for transmitting light. Optical fibres are usually thinner than a human hair! The construction of a single optical fibre is shown in Figure 5.17.

The basic functional structure of an optical fibre consists of an outer protective *cladding* and an *inner core* through which light pulses travel. The overall diameter of the fibre is about $125\ \mu\text{m}$ ($125 \times 10^{-6}\ \text{m}$) and that of the core is just about $50\ \mu\text{m}$ ($50 \times 10^{-6}\ \text{m}$). The difference in refractive index of the cladding and the core allows total internal reflection to occur in the same way as happens at an air-water surface. If light is incident on a cable end with an angle of incidence greater than the critical angle then the light will remain trapped inside the glass strand. In this way, light travels very quickly down the length of the cable.

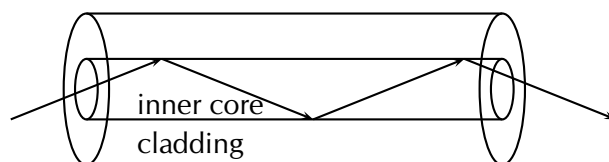


Figure 5.17: Structure of a single optical fibre.

FACT

Endoscopy means to *look inside* and refers to looking inside the human body for diagnosing medical conditions.

Fibre Optics in Telecommunications Optical fibres are most common in telecommunications, because information can be transported over long distances, with minimal loss of data. This gives optical fibres an advantage over conventional cables.

Signals are transmitted from one end of the fibre to another in the form of laser pulses. A single strand of fibre optic cable is capable of handling over 3000 transmissions at the same time which is a huge improvement over the conventional co-axial cables. Multiple signal transmission is achieved by sending individual light pulses at slightly different angles. For example if one of the pulses makes a $72,23^\circ$ angle of incidence then a separate pulse can be sent at an angle of $72,26^\circ$! The transmitted signal is received almost instantaneously at the other end of the cable since the information coded onto the laser travels at the speed of light! During transmission over long distances *repeater stations* are used to amplify the signal which has weakened by the time it reaches the station. The amplified signals are then relayed towards their destination and may encounter several other repeater stations on the way.

Fibre optics in medicine Optic fibres are used in medicine in *endoscopes*.

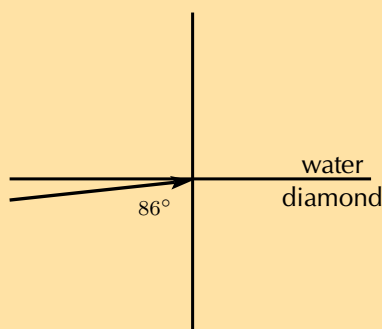
The main part of an endoscope is the optical fibre. Light is shone down the optical fibre and a medical doctor can use the endoscope to look inside the body of a patient. Endoscopes can be used to examine the inside of a patient's stomach, by inserting the endoscope down the patient's throat.

Endoscopes also allow minimally invasive surgery. This means that a person can be diagnosed and treated through a small incision (cut). This has advantages over open surgery because endoscopy is quicker and cheaper and the patient recovers more quickly. The alternative is open surgery which is expensive, requires more time and is more traumatic for the patient.

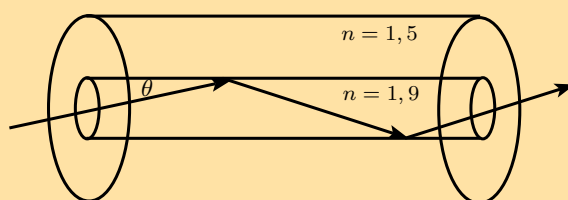
Exercise 5 – 5: Total internal reflection and fibre optics

1. Describe total internal reflection by using a diagram and referring to the conditions that must be satisfied for total internal reflection to occur.
2. Define what is meant by the *critical angle* when referring to total internal reflection. Include a ray diagram to explain the concept.
3. Will light travelling from diamond to silicon ever undergo total internal reflection?
4. Will light travelling from sapphire to diamond undergo total internal reflection?
5. What is the critical angle for light travelling from air to acetone?

6. Light travelling from diamond to water strikes the interface with an angle of incidence of 86° as shown in the picture. Calculate the critical angle to determine whether the light be totally internally reflected and so be trapped within the diamond.



7. Which of the following interfaces will have the largest critical angle?
- a glass to water interface
 - a diamond to water interface
 - a diamond to glass interface
8. If a fibre optic strand is made from glass, determine the critical angle of the light ray so that the ray stays within the fibre optic strand.
9. A glass slab is inserted in a tank of water. If the refractive index of water is 1,33 and that of glass is 1,5, find the critical angle.
10. A diamond ring is placed in a container full of glycerin. If the critical angle is found to be $37,4^\circ$ and the refractive index of glycerin is given to be 1,47, find the refractive index of diamond.
11. An optical fibre is made up of a core of refractive index 1,9, while the refractive index of the cladding is 1,5. Calculate the maximum angle which a light pulse can make with the wall of the core. NOTE: The question does not ask for the angle of incidence but for the angle made by the ray with the wall of the core, which will be equal to $90^\circ - \text{angle of incidence}$.



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- Light rays are lines which are perpendicular to the light's wavefronts. In geometrical optics we represent light rays with arrows with straight lines.
- Light rays reflect off surfaces. The incident ray shines in on the surface and the reflected ray is the one that bounces off the surface. The normal is the line perpendicular to the surface where the light strikes the surface.
- The angle of incidence is the angle between the incident ray and the surface, and the incident ray, reflected ray, and the normal, all lie in the same plane.
- The Law of Reflection states the angle of incidence (θ_i) is equal to the angle of reflection (θ_r) and that the reflected ray lies in the plane of incidence.
- Light can be absorbed and transmitted.
- The speed of light, c , is constant in a given medium and has a maximum speed in vacuum of $3 \times 10^8 \text{ m}\cdot\text{s}^{-1}$
- Refraction occurs at the boundary of two media when light travels from one medium into the other and its speed changes but its frequency remains the same. If the light ray hits the boundary at an angle which is not perpendicular to or parallel to the surface, then it will change direction and appear to 'bend'.
- The refractive index (symbol n) of a material is the ratio of the speed of light in a vacuum to its speed in the material and gives an indication of how difficult it is for light to get through the material.

$$n = \frac{c}{v}$$

- Optical density is a measure of the refracting power of a medium.
- The normal to a surface is the line which is perpendicular to the plane of the surface.
- The angle of incidence is the angle defined between the normal to a surface and the incoming (incident) light ray.
- The angle of refraction is the angle defined between the normal to a surface and the refracted light ray.
- Snell's Law gives the relationship between the refractive indices, angles of incidence and reflection of two media.

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

- Light travelling from one medium to another of lighter optical density will be refracted towards the normal.

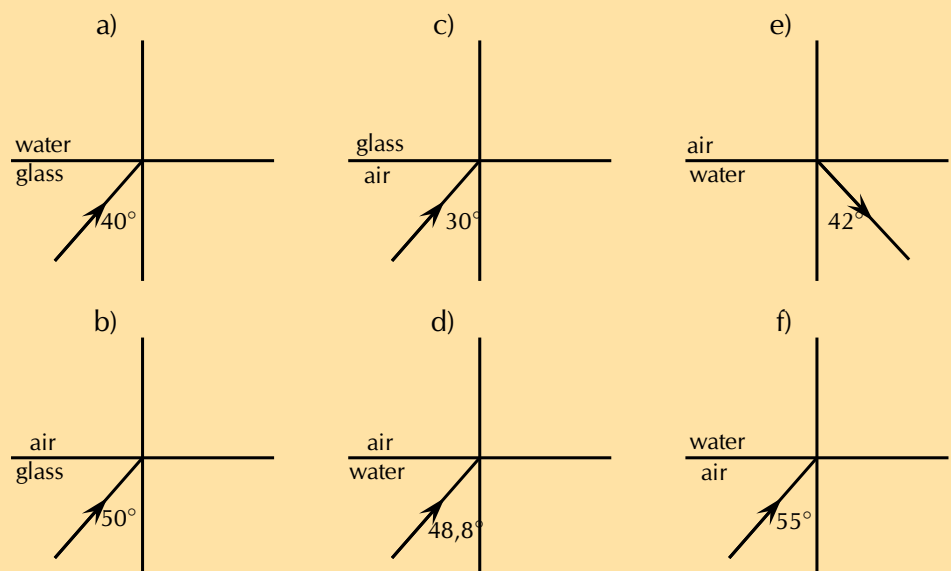
Light travelling from one medium to another of lower optical density will be refracted away from the normal.

- The critical angle of a medium is the angle of incidence when the angle of refraction is 90° and the refracted ray runs along the interface between the two media.

- Total internal reflection takes place when light travels from one medium to another of lower optical density. If the angle of incidence is greater than the critical angle for the medium, the light will be reflected back into the medium. No refraction takes place.
- Total internal reflection is used in optical fibres in telecommunication and in medicine in endoscopes. Optical fibres transmit information much more quickly and accurately than traditional methods.

Exercise 5 – 6: End of chapter exercises

1. Give one word for each of the following descriptions:
 - a) The perpendicular line that is drawn at right angles to a reflecting surface at the point of incidence.
 - b) The bending of light as it travels from one medium to another.
 - c) The bouncing of light off a surface.
2. State whether the following statements are **true** or **false**. If they are false, rewrite the statement correcting it.
 - a) The refractive index of a medium is an indication of how fast light will travel through the medium.
 - b) Total internal refraction takes place when the incident angle is larger than the critical angle.
 - c) The speed of light in a vacuum is about $3 \times 10^8 \text{ m}\cdot\text{s}^{-1}$.
3. Complete the following ray diagrams to show the path of light.



4. A ray of light strikes a surface at 35° to the normal to the surface. Draw a ray diagram showing the incident ray, reflected ray and surface normal. Calculate the angles of incidence and reflection and fill them in on your diagram.
5. Light travels from glass ($n = 1,5$) to acetone ($n = 1,36$). The angle of incidence is 25° .

- a) Describe the path of light as it moves into the acetone.
 - b) Calculate the angle of refraction.
 - c) What happens to the speed of the light as it moves from the glass to the acetone?
 - d) What happens to the wavelength of the light as it moves into the acetone?
6. Light strikes the interface between diamond and an unknown medium with an incident angle of 32° . The angle of refraction is measured to be 46° . Calculate the refractive index of the medium and identify the medium.
 7. Explain what total internal reflection is and how it is used in medicine and telecommunications. Why is this technology much better to use?

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2D and 3D wavefronts

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6.1 Introduction

ESBND

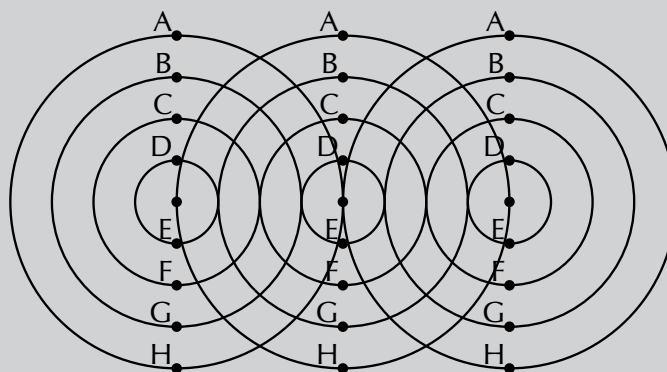
You have learnt about the basic properties of waves before, specifically about reflection and refraction. In this chapter, you will learn about phenomena that arise with waves in two and three dimensions: diffraction. We will also build on interference which you have learnt about previously but now in more than one dimension.

6.2 Wavefronts

ESBNF

Investigation: Wavefronts

The diagram below shows three identical waves being emitted by three point sources. A point source is something that generates waves and is so small that we consider it to be a point. It is not large enough to affect the waves. All points marked with the same letter are in phase. Join all points with the same letter.



What type of lines (straight, curved, etc.) do you get? How does this compare to the line that joins the sources?

Consider three point sources of waves. If each source emits waves isotropically (i.e. the same in all directions) we will get the situation shown in as shown in Figure 6.1 below.

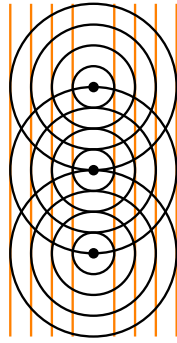


Figure 6.1:

Wavefronts are imaginary lines joining waves that are in phase. In the example, the wavefronts (shown by the orange, vertical lines) join all waves at the crest of their cycle.

We define a **wavefront** as the imaginary line that joins waves that are in phase. These are indicated by the orange, vertical lines in Figure 6.1. The points that are in phase can be peaks, troughs or anything in between, it doesn't matter which points you choose as long as they are in phase.

6.3 Huygens principle

ESBNG

Christiaan Huygens described how to determine the path of waves through a medium.

DEFINITION: The Huygens Principle

Every point of a wave front serves as a point source of spherical, secondary waves. After a time t , the new position of the wave front will be that of a surface tangent to the secondary waves.

Huygens principle applies to any wavefront, even those that are curved as you would get from a single point source. A simple example of the Huygens Principle is to consider the single wavefront in Figure 6.2.

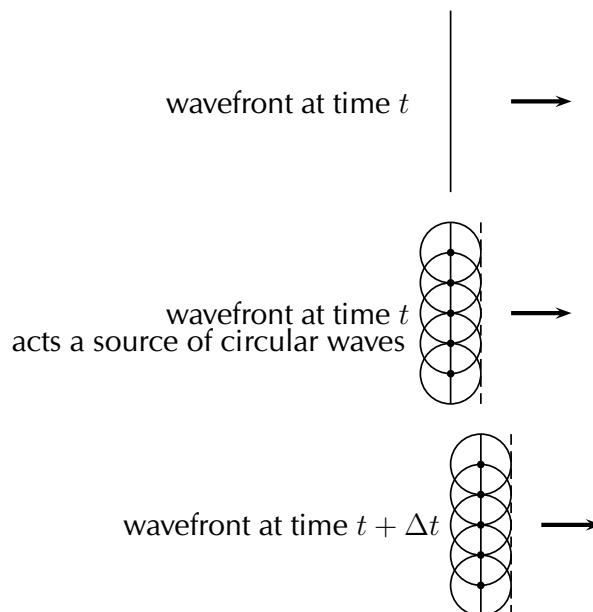
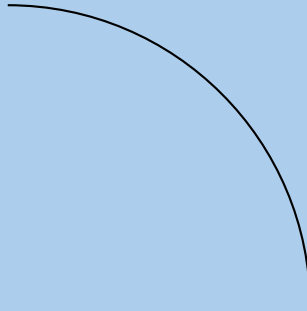


Figure 6.2: 'n Enkele golfvront by tydstep t tree op soos 'n versameling puntbronne van sirkelvormige golwe wat mekaar steur om 'n nuwe golfvront by 'n tydstep $t + \Delta t$. te gee. Die proses hou aan en is van toepassing op enige tipe golfvorm

Worked example 1: Application of the Huygens principle

QUESTION

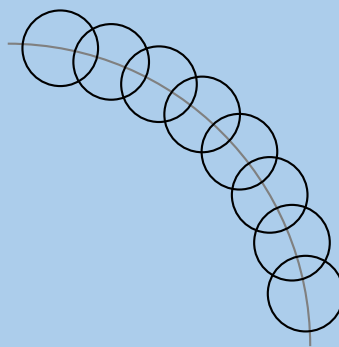
Given the wavefront,



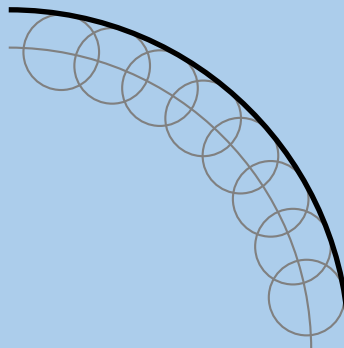
use the Huygens Principle to determine the wavefront at a later time.

SOLUTION

Step 1: Draw circles at various points along the given wavefront



Step 2: Join the circle crests to get the wavefront at a later time

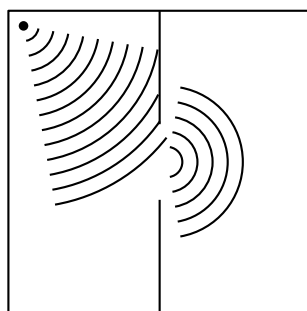


One of the most interesting, and also very useful, properties of waves is **diffraction**.

DEFINITION: *Diffraction*

Diffraction is the ability of a wave to spread out in wavefronts as the wave passes through a small aperture or around a sharp edge.

For example, if two rooms are connected by an open doorway and a sound is produced in a remote corner of one of them, a person in the other room will hear the sound as if it originated at the doorway.



As far as the second room is concerned, the vibrating air in the doorway is the source of the sound.

This means that when waves move through small holes they appear to bend around the sides because there are not enough points on the wavefront to form another straight wavefront. This is bending round the sides we call *diffraction*.

Extension

Diffraction

Diffraction effects are more clear for water waves with longer wavelengths. Diffraction can be demonstrated by placing small barriers and obstacles in a ripple tank and observing the path of the water waves as they encounter the obstacles. The waves are seen to pass around the barrier into the regions behind it; subsequently the water behind the barrier is disturbed. The amount of diffraction (the sharpness of the bending) increases with increasing wavelength and decreases with decreasing wavelength. In fact, when the wavelength of the waves are smaller than the obstacle, no noticeable diffraction occurs.

This experiment demonstrates diffraction using water waves in a ripple tank. You can also demonstrate diffraction using a single slit and a light source with coloured filters.

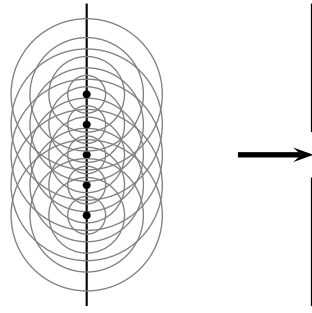
Water waves in a ripple tank can be used to demonstrate diffraction and interference.

- Turn on the wave generator so that it produces waves with a high frequency (short wavelength).
 - Place a few obstacles, one at a time, (e.g. a brick or a ruler) in the ripple tank. What happens to the wavefronts as they propagate near/past the obstacles? Draw your observations.
 - How does the diffraction change when you change the size of the object?
- Now turn down the frequency of the wave generator so that it produces waves with longer wavelengths.
 - Place the same obstacles in the ripple tank (one at a time). What happens to the wavefronts as they propagate near/past the obstacles? Draw your observations.
 - How does the diffraction change from the higher frequency case?
- Remove all obstacles from the ripple tank and insert a second wave generator. Turn on both generators so that they start at the same time and have the same frequency.
 - What do you notice when the two sets of wavefronts meet each other?
 - Can you identify regions of constructive and destructive interference?
- Now turn on the generators so that they are out of phase (i.e. start them so that they do not make waves at exactly the same time).
 - What do you notice when the two sets of wavefronts meet each other?
 - Can you identify regions of constructive and destructive interference?

6.5 Diffraction through a single slit

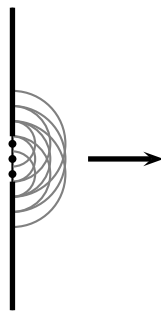
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Waves diffract when they encounter obstacles. Why does this happen? If we apply Huygens principle it becomes clear. Think about a wavefront impinging on a barrier with a slit in it, only the points on the wavefront that move into the slit can continue emitting forward moving waves - but because a lot of the wavefront has been blocked by the barrier, the points on the edges of the hole emit waves that bend round the edges. How to use this approach to understand what happens is sketched below:



Before the wavefront strikes the barrier the wavefront generates another forward moving wavefront (applying Huygens' principle). Once the barrier blocks most of the wavefront you can see that the forward moving wavefront bends around the slit because the secondary waves they would need to interfere with to create a straight wavefront have been blocked by the barrier.

If you employ Huygens' principle you can see the effect is that the wavefronts are no longer straight lines.



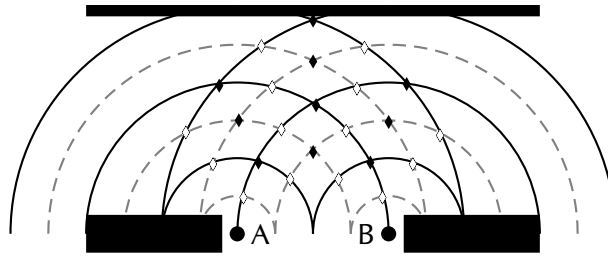
Diffraction patterns

ESBNK

We can learn even more about what happens after the wavefront strikes the barrier by applying Huygens' principle further.

Each point on the wavefront moving through the slit acts like a point source. We can think about some of the effects of this if we analyse what happens when two point sources are close together and emit wavefronts with the same wavelength and frequency. These two point sources represent the point sources on the two edges of the slit and we can call the source A and source B.

Each point source emits wavefronts from the edge of the slit. In the diagram we show a series of wavefronts emitted from each point source. The black lines show peaks in the waves emitted by the point sources and the gray lines represent troughs. We label the places where constructive interference (peak meets a peak or trough meets a trough) takes place with a solid diamond and places where destructive interference (trough meets a peak) takes place with a hollow diamond. When the wavefronts hit a barrier there will be places on the barrier where constructive interference takes place and places where destructive interference happens.

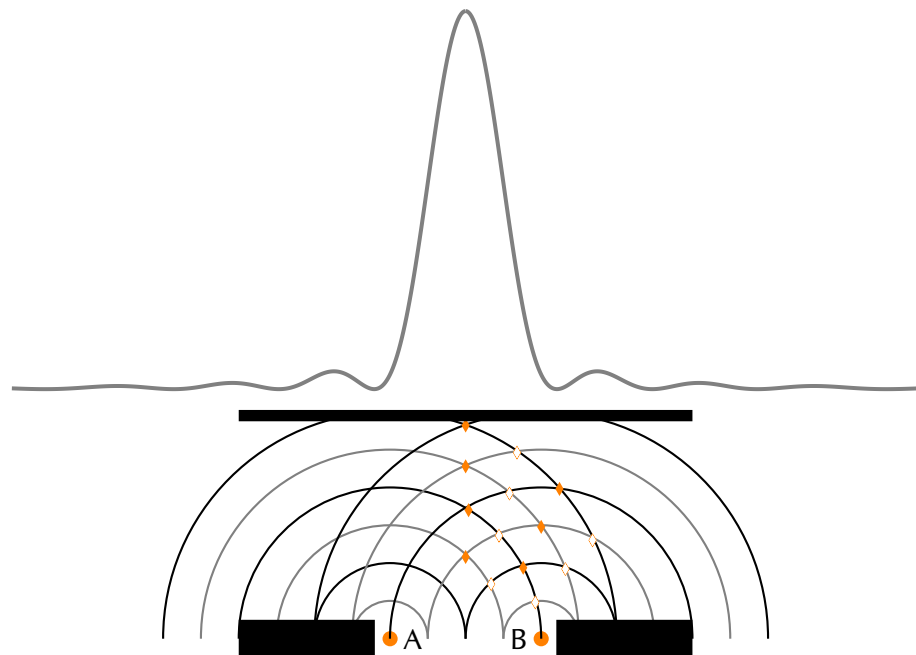


The measurable effect of the constructive or destructive interference at a barrier depends on what type of waves we are dealing with. If we were dealing with sound waves, then it would be very noisy at points along the barrier where the constructive interference is taking place and quiet where the destructive interference is taking place.

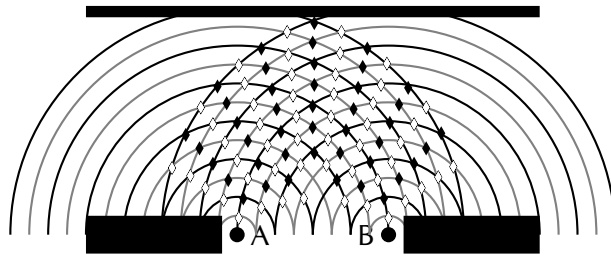
NOTE:

The pattern of constructive then destructive interference measured some distance away from a single slit is caused because of two properties of waves, diffraction **and** interference. Sometimes this pattern is called an interference pattern and sometimes it is called a diffraction pattern. Both names are correct and both properties are required for the pattern to be observed. For consistency we will call it a *diffraction pattern* in for the rest of this book.

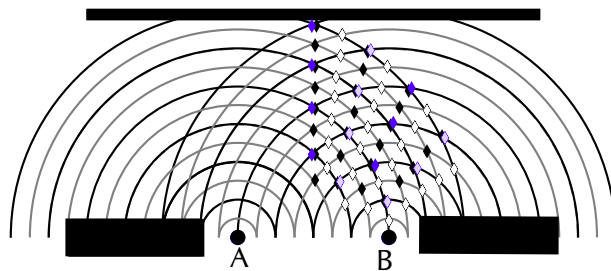
The intensity of the diffraction pattern for a single narrow slit looks like this:



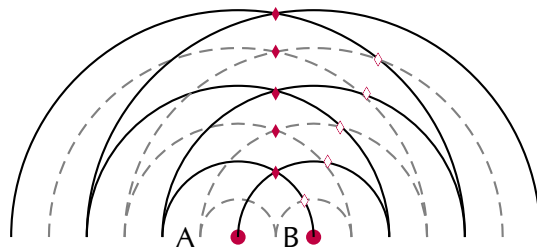
The picture above sketches how the wavefronts interfere to form the diffraction pattern. The peaks correspond to places where the waves are adding constructively and the minima are places where destructive interference is taking place. If you look at the picture you can see that if the wavelength (the distance between two consecutive peaks/troughs) of the waves were different the pattern would be different. For example, if the wavelength were halved the sketch would be:



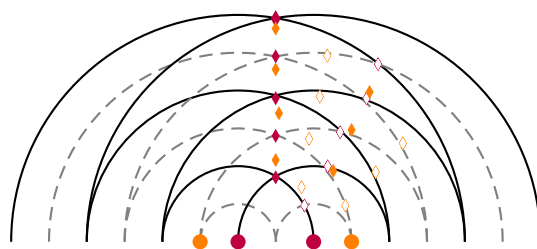
The amount that the waves diffract depends on the wavelength. We can compare the spread in the points of constructive and destructive interference by plotting the highlighted points together for the two cases. We have to line up the central maximum from the two cases to see the difference. The case where the wavelength is smaller results in smaller angles between the lines of constructive and destructive interference.



It also depends on the width of the slit, changing the width of the slit would change the distance between the points labelled A and B in the sketch. For example, if we repeat the sketch halving the distance between the points A and B we would get:



We can compare the spread in the points of constructive and destructive interference by plotting the highlighted points together for the two cases. We have to line up the central maximum from the two cases to see the difference. The case where the two points are closer together, in purple, results in bigger angles between the lines of constructive and destructive interference.



Using our sketches we see that the extent to which the diffracted wave passing through the slit spreads out depends on the width of the slit and the wavelength of the waves. The narrower the slit, the more diffraction there is and the shorter the wavelength the less diffraction there is. The degree to which diffraction occurs is:

$$\text{diffraction} \propto \frac{\lambda}{w}$$

where λ is the wavelength of the wave and w is the width of the slit.

We can do a sanity check on the relationship by considering some special cases, very big and very small values for each of the numerator and denominator to see what sort of behaviour we expect (this is not a calculation, just a check to see what sort outcomes we expect when we change wavelength or slit width):

- Set $\lambda = 1$ and w very large, the result will be $\frac{1}{\text{very big number}}$ which is a very small number. So for a very big slit there is very little diffraction.
- Set $\lambda = 1$ and w very small, the result will be $\frac{1}{\text{very small number}}$ which is a very big number. So for a very small slit there is large diffraction (this makes sense because eventually you are dealing with a point source which emits circular wavefronts).
- Set λ very large and $w = 1$, the result will be $\frac{\text{very big number}}{1}$ which is a very big number. So for a very big wavelength there is large diffraction.
- Set λ very small and $w = 1$, the result will be $\frac{\text{very small number}}{1}$ which is a very small number. So for a very small wavelength there is little diffraction.

Wave nature of light

In Grade 10 we learnt about electromagnetic radiation and that visible light is a small part of the EM spectrum. EM radiation is a wave so we should see diffraction for visible light when it strikes a barrier or passes through a slit. In everyday life you don't notice diffraction of light around objects or when light passes through an open door or window. This is because the wavelength of light is very small and the "slits" like doors and windows are quite large.

We can put some everyday numbers into

$$\text{diffraction} \propto \frac{\lambda}{w}$$

to see how much diffraction we expect. White light is combination of light of many different colours and each colour has a different frequency or wavelength. To make things simpler let's just think about one colour, green light has a wavelength of 532×10^{-9} m. If a wavefront of green light struck the wall of a house with an open door that is 1 m wide what would we expect to see?

$$\begin{aligned} \text{diffraction} &\propto \frac{\lambda}{w} \\ &\propto \frac{532 \times 10^{-9} \text{ m}}{1 \text{ m}} \\ &\propto 532 \times 10^{-9} \end{aligned}$$



Figure 6.3: A diffraction grating reflecting green light.

The result is a very small number so we expect to see very little diffraction. In fact, the effect is so small that we cannot see it with the human eye. We can observe diffraction of green light but for us to get diffraction $\propto 1$ we need the wavelength and slit width to be the same number. So we know the effects of diffraction should become noticeable when the wavelength and slit width are similar. We can't change the wavelength of green light but there are objects called diffraction gratings that have very narrow slits that we can use to study the diffraction of light. We let wavefronts of green light strike a diffraction grating and then put a screen on the other side. We can see where the intensity of the light on the screen is large and where it is small. For green light on a particular diffraction grating the pattern of green light on the screen looks like:



Blue light with a wavelength of 450×10^{-9} m and the same diffraction grating will produce:



Worked example 2: Diffraction

QUESTION

Two diffraction patterns are presented, determine which one has the longer wavelength based on the features of the diffraction pattern. The first pattern is for green light:



The second pattern is for red light:



The same diffraction grating is used in to generate both diffraction patterns.

SOLUTION

Step 1: Determine what is required

We need to compare the diffraction patterns to extract information about the relative wavelengths so we can decide which one is longer. We know that the diffraction pattern depends on wavelength and slit width through:

$$\text{diffraction} \propto \frac{\lambda}{w}$$

The diffraction grating is the same in both cases so we know that the slit width is fixed.

Step 2: Analyse patterns

By eye we can see that the red pattern is wider than the green pattern. There is more diffraction for the red light, this means that:

$$\text{diffraction}_{\text{red}} > \text{diffraction}_{\text{green}}$$

$$\frac{\lambda_{\text{red}}}{w} > \frac{\lambda_{\text{green}}}{w}$$

$$\lambda_{\text{red}} > \lambda_{\text{green}}$$

Step 3: Final answer

The wavelength of the red light is longer than that of the green light.

Exercise 6 – 1:

1. As the width of the slit producing a single-slit diffraction pattern is reduced, how will the diffraction pattern produced change?
2. A water break at the entrance to a harbour consists of a rock barrier with a 50 m wide opening. Ocean waves of 20 m wavelength approach the opening straight on. Light with a wavelength of 500×10^{-9} m strikes a single slit of width 30×10^{-9} m. Which waves are diffracted to a greater extent?
3. For the diffraction pattern below, sketch what you expect to change if:



- a) the wavelength gets larger
- b) the wavelength gets smaller
- c) the slit width gets larger
- d) the slit width gets smaller
- e) the frequency of the wave gets smaller
- f) the frequency of the wave gets larger

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3e. 23VH 3f. 23VJ



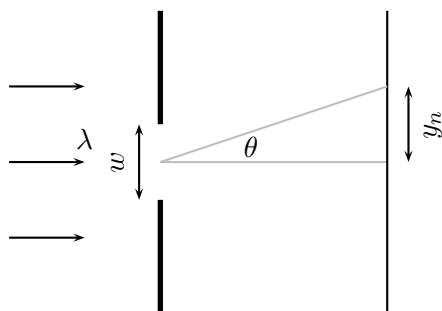
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This section is for interest and extension, **skip to end of chapter summary if you are following CAPS.**

There is a formula we can use to determine where the peaks and minima are in the interference spectrum. There will be more than one minimum. There are the same number of minima on either side of the central peak and the distances from the first one on each side are the same to the peak. The distances to the peak from the second minimum on each side is also the same, in fact the two sides are mirror images of each other. We label the first minimum that corresponds to a positive angle from the centre as $m = 1$ and the first on the other side (a negative angle from the centre) as $m = -1$, the second set of minima are labelled $m = 2$ and $m = -2$ etc.



The equation for the angle at which the minima occur is given in the definition below:

DEFINITION: *Interference Minima*

The angle at which the minima in the interference spectrum occur is:

$$\sin \theta = \frac{m\lambda}{w}$$

where

θ is the angle to the minimum

w is the width of the slit

λ is the wavelength of the impinging wavefronts

m is the order of the minimum, $m = \pm 1, \pm 2, \pm 3, \dots$

Worked example 3: Diffraction minimum

QUESTION

A slit with a width of 2511 nm has red light of wavelength 650 nm impinge on it. The diffracted light interferes on a surface. At which angle will the first minimum be?

SOLUTION

Step 1: Check what you are given

We know that we are dealing with diffraction patterns from the diffraction of light passing through a slit. The slit has a width of 2511 nm which is 2511×10^{-9} m and

we know that the wavelength of the light is 650 nm which is 650×10^{-9} m. We are looking to determine the angle to first minimum so we know that $m = 1$.

Step 2: Applicable principles

We know that there is a relationship between the slit width, wavelength and interference minimum angles: $\sin \theta = \frac{m\lambda}{w}$

We can use this relationship to find the angle to the minimum by substituting what we know and solving for the angle.

Step 3: Substitution

$$\begin{aligned}\sin \theta &= \frac{650 \times 10^{-9} \text{ m}}{2511 \times 10^{-9} \text{ m}} \\ \sin \theta &= \frac{650}{2511} \\ \sin \theta &= 0,258861012 \\ \theta &= \sin^{-1} 0,258861012 \\ \theta &= 15^\circ\end{aligned}$$

The first minimum is at 15° from the centre maximum.

Worked example 4: Diffraction minimum

QUESTION

A slit with a width of 2511 nm has green light of wavelength 532 nm impinge on it. The diffracted light interferes on a surface, at what angle will the first minimum be?

SOLUTION

Step 1: Check what you are given

We know that we are dealing with diffraction patterns from the diffraction of light passing through a slit. The slit has a width of 2511 nm which is 2511×10^{-9} m and we know that the wavelength of the light is 532 nm which is 532×10^{-9} m. We are looking to determine the angle to first minimum so we know that $m = 1$.

Step 2: Applicable principles

We know that there is a relationship between the slit width, wavelength and interference minimum angles: $\sin \theta = \frac{m\lambda}{w}$

We can use this relationship to find the angle to the minimum by substituting what we know and solving for the angle.

Step 3: Substitution

$$\begin{aligned}\sin \theta &= \frac{532 \times 10^{-9} \text{ m}}{2511 \times 10^{-9} \text{ m}} \\ \sin \theta &= \frac{532}{2511} \\ \sin \theta &= 0,211867782 \\ \theta &= \sin^{-1} 0,211867782 \\ \theta &= 12.2^\circ\end{aligned}$$

The first minimum is at $12,2^\circ$ from the centre peak.

From the formula $\sin \theta = \frac{m\lambda}{w}$ you can see that a smaller wavelength for the same slit results in a smaller angle to the interference minimum. This is something you just saw in the two worked examples. Do a sanity check, go back and see if the answer makes sense. Ask yourself which light had the longer wavelength, which light had the larger angle and what do you expect for longer wavelengths from the formula.

Worked example 5: Diffraction minimum

QUESTION

A slit has a width which is unknown and has green light of wavelength 532 nm impinge on it. The diffracted light interferes on a surface, and the first minimum is measure at an angle of $20,77^\circ$?

SOLUTION

Step 1: Check what you are given

We know that we are dealing with diffraction patterns from the diffraction of light passing through a slit. We know that the wavelength of the light is 532 nm which is $532 \times 10^{-9} \text{ m}$. We know the angle to first minimum so we know that $m = 1$ and $\theta = 20.77^\circ$.

Step 2: Applicable principles

We know that there is a relationship between the slit width, wavelength and interference minimum angles: $\sin \theta = \frac{m\lambda}{w}$

We can use this relationship to find the width by substituting what we know and solving for the width.

Step 3: Substitution

$$\sin \theta = \frac{532 \times 10^{-9} \text{ m}}{w}$$

$$\sin 20,77^\circ = \frac{532 \times 10^{-9}}{w}$$

$$w = \frac{532 \times 10^{-9}}{0,3546666667}$$

$$w = 1500 \times 10^{-9}$$

$$w = 1500 \text{ nm}$$

The slit width is 1500 nm.

► See simulation: [23VK](#) at www.everythingscience.co.za

6.6 Chapter summary

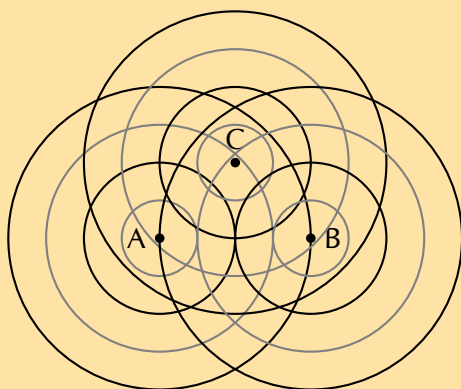
ESBNQ

► See presentation: [23VM](#) at www.everythingscience.co.za

- A wavefront is an imaginary line that connects waves that are in phase.
- Huygen's Principle states that every point of a wave front serves as a point source of spherical, secondary waves. After a time t , the new position of the wave front will be that of a surface tangent to the secondary waves.
- Diffraction is the ability of a wave to spread out in wavefronts as the wave passes through a small aperture or around a sharp edge.
- When a wave passes through a slit, diffraction of the wave occurs. Diffraction of the wave is when the wavefront spreads out or "bends" around corners.
- The degree of diffraction depends on the width of the slit and the wavelength of the wave with: diffraction $\propto \frac{\lambda}{w}$ where λ is the wavelength of the wave and w is the width of the slit.

Exercise 6 – 2:

1. In the diagram below the peaks of wavefronts are shown by black lines and the troughs by grey lines. Mark all the points where constructive interference between two waves is taking place and where destructive interference is taking place. Also note whether the interference results in a peak or a trough.



2. For a slit of width 1300 nm, order the following EM waves from least to most diffracted:
 - a) green at 510 nm
 - b) blue at 475 nm
 - c) red at 650 nm
 - d) yellow at 570 nm

3. For light of wavelength 540 nm, determine which of the following slits widths results in the maximum and which results in the minimum amount of diffraction
 - a) 323×10^{-9} m
 - b) 12,47 nm
 - c) 21,1 pm

4. For light of wavelength 635 nm, determine what the width of the slit needs to be to have the diffraction be less than the angle of diffraction in each of these cases:
 - a) Water waves at the entrance to a harbour which has a rock barrier with a 3 m wide opening. The waves have a wavelength of 16 m wavelength approach the opening straight on.
 - b) Light with a wavelength of 786×10^{-9} m strikes a single slit of width 30×10^{-7} m.

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1. 23VN 2. 23VP 3. 23VQ 4. 23VR



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Ideal gases

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We are surrounded by gases in our atmosphere which support and protect life on this planet. Everyday we breathe in oxygen and release carbon dioxide. Green plants take in the carbon dioxide and release oxygen. One way or another we are surrounded by a mix of many different gases, some that we need and some that are harmful to us. In this chapter, we are going to learn more about gases and about the different gas laws.



Key Mathematics Concepts

- Ratio and proportion — Physical Sciences, Grade 10, Science skills
- Equations — Mathematics, Grade 10, Equations and inequalities
- Graphs — Mathematics, Grade 10, Functions and graphs
- Units and unit conversions — Physical Sciences, Grade 10, Science skills

7.1 Motion of particles

ESBNR

The kinetic theory of gases

ESBNS

In grade 10 you learnt about the kinetic theory of matter. The kinetic theory of matter says that all matter is composed of particles which have a certain amount of energy which allows them to move at different speeds depending on the temperature (energy). There are spaces between the particles and also attractive forces between particles when they come close together.

Now we will look at applying the same ideas to gases.

The main assumptions of the kinetic theory of gases are as follows:

- Gases are made up of **particles** (e.g. atoms or molecules). The size of these particles is very small compared to the distance between the particles.
- These particles are constantly moving because they have kinetic energy. The particles move in straight lines at different speeds.
- There are **attractive forces** between particles. These forces are very weak for gases.
- The collisions between particles and the walls of the container do not change the kinetic energy of the system.
- The **temperature** of a gas is a measure of the **average kinetic energy** of the particles.

From these assumptions we can define the pressure and temperature of any gas.

DEFINITION: *Pressure*

The **pressure** of a gas is a measure of the number of collisions of the gas particles with each other and with the sides of the container that they are in.

▶ See video: 23VS at www.everythingscience.co.za

DEFINITION: *Temperature*

The **temperature** of a substance is a measure of the **average kinetic energy** of the particles.

If the gas is heated (i.e. the temperature increases), the average kinetic energy of the gas particles will increase and if the temperature is decreased, the average kinetic energy of the particles decreases. If the energy of the particles decreases significantly, the gas liquefies (becomes a liquid).

One of the assumptions of the kinetic theory of gases is that all particles have a different speed. However, this is only the case for a **real** gas. For an ideal gas we assume that all particles in the gas have the **same** speed.

So for an ideal gas we can simply talk about the speed of particles. But for a real gas we must use the average speed of all the particles.

▶ See video: 23VT at www.everythingscience.co.za

Ideal gases and non-ideal gas behaviour

ESBNT

When we look at the gas laws in the next section we will only deal with ideal gases.

DEFINITION: *Ideal gas*

An ideal gas has identical particles of zero volume, with no intermolecular forces between them. The atoms or molecules in an ideal gas move at the same speed.

Almost all gases obey the gas laws within a limited range of pressures and temperatures. So we can use the gas laws to predict how real gases will behave.

DEFINITION: *Real gas*

Real gases behave more or less like ideal gases except at high pressures and low temperatures.

Before we go on to look at the gas laws we will first see what happens to gases at high pressures and low temperatures.

When we defined an ideal gas, we said that an ideal gas has identical particles of

zero volume and that there are no intermolecular forces between the particles in the gas. We need to look more closely at these statements because they affect how gases behave at high pressures or at low temperatures.

1. Molecules do occupy volume

When pressures are very high and the molecules are compressed, the volume of the molecules becomes significant. This means that the total volume available for the gas molecules to move is reduced and collisions become more frequent. This causes the pressure of the gas to be *higher* than what would be expected for an ideal gas (Figure 7.1).

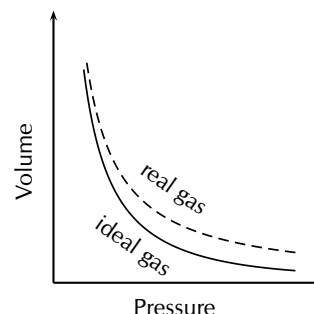


Figure 7.1:
Gases deviate from ideal gas behaviour at high pressure.

2. Forces of attraction do exist between molecules

At low temperatures, when the speed of the molecules decreases and they move closer together, the intermolecular forces become more apparent. As the attraction between molecules increases, their movement decreases and there are fewer collisions between them. The pressure of the gas at low temperatures is therefore lower than what would have been expected for an ideal gas (Figure 7.2). If the temperature is low enough or the pressure high enough, a real gas will **liquefy**.

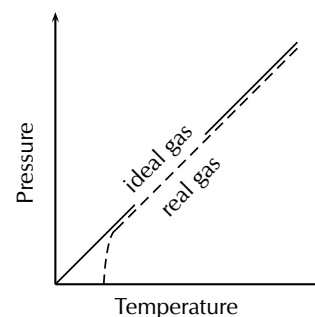


Figure 7.2:
Gases deviate from ideal gas behaviour at low temperatures.

Exercise 7 – 1:

- Summarise the difference between a real gas and an ideal gas in the following table:

Property	Ideal gas	Real gas
Size of particles		
Attractive forces		
Speed of molecules		

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1. 23VV



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There are several laws to explain the behaviour of ideal gases. The first three that we will look at apply under very strict conditions. These laws are then combined to form the general gas equation and the ideal gas equation.

Before we start looking at these laws we need to look at some common conversions for units.

The following table gives the SI units. This table also shows how to convert between common units. Do not worry if some of the units are strange to you. By the end of this chapter you will have had a chance to see all these units in action.

Variable	SI Units	Other units
Pressure (p)	Pascals (Pa)	760 mm Hg = 1 atm = 101 325 Pa = 101,325 kPa
Volume (V)	m ³	1 m ³ = 1 000 000 cm ³ = 1000 dm ³ = 1000 L
Moles (n)	mol	
Universal gas constant (R)	J·K ⁻¹ ·mol ⁻¹	
Temperature (K)	Kelvin (K)	K = °C + 273

Table 7.1: Conversion table showing SI units of measurement and common conversions.

Two very useful volume relations to remember are: 1 mL = 1 cm³ and 1 L = 1 dm³.

Boyle's law: Pressure and volume of an enclosed gas ESBNW

If you have ever tried to force in the plunger of a syringe or a bicycle pump while sealing the opening with your finger, you will have seen Boyle's Law in action! The following experiment will allow you to see this law in action.

DEFINITION: Boyle's Law

The pressure of a fixed quantity of gas is inversely proportional to the volume it occupies so long as the temperature remains constant.

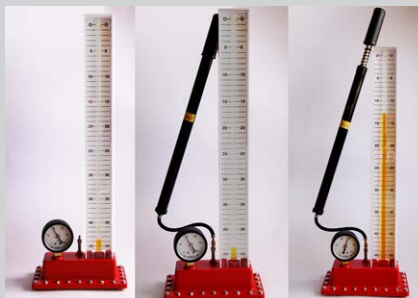
Informal experiment: Boyle's law

Aim:

To verify Boyle's law.

Apparatus:

Boyle's law apparatus (a syringe or bicycle pump attached to a pressure gauge.)



Method:

1. Use the pump to fill the syringe (or glass tube) until the pressure gauge reads the maximum value. Note the volume and the pressure reading.
2. Slowly release some of the air until the pressure has dropped by about 20 units (the units will depend on what your pressure gauge measures, e.g. kPa).
3. Let the system stabilise for about 2 min and then read the volume.
4. Repeat the above two steps until you have six pairs of pressure-volume readings.

Results:

Record your results in the following table. Remember that your pressure and volume units will be determined by the apparatus you are using.

Pressure	Volume

What happens to the volume as the pressure decreases?

Plot your results as a graph of pressure versus volume (in other words plot pressure on the x-axis and volume on the y-axis). Pressure is the independent variable, which we are changing to see what happens to volume.

Plot your results as a graph of pressure versus the inverse of volume (in other words $\frac{1}{V}$, so for each volume reading you will work out the value of 1 divided by the volume reading).

What do you notice about each graph?

Conclusion:

If the volume of the gas decreases, the pressure of the gas increases. If the volume of the gas increases, the pressure decreases. These results support Boyle's law.

In the above experiment, the volume of the gas decreased when the pressure increased, and the volume increased when the pressure decreased. This is called an **inverse relationship** (or more-less relationship). The inverse relationship between pressure and volume is shown in Figure 7.3.

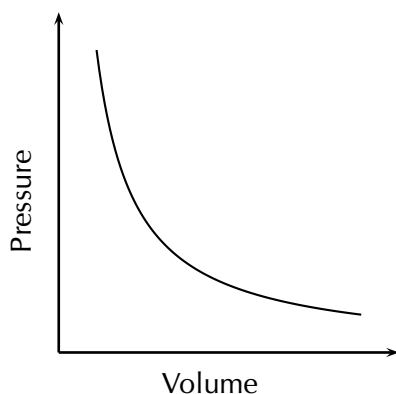
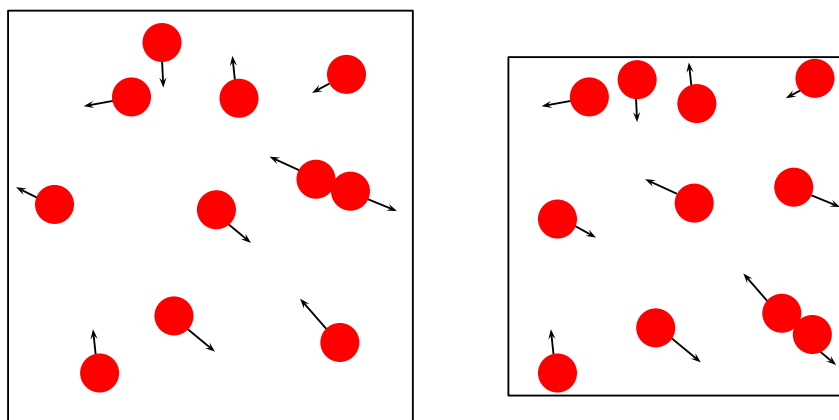


Figure 7.3: Graph showing the inverse relationship between pressure and volume.

Can you use the kinetic theory of gases to explain this inverse relationship between the pressure and volume of a gas? Let's think about it. If you decrease the volume of a gas, this means that the same number of gas particles are now going to come into contact with each other and with the sides of the container much more often. But we said that **pressure** is a measure of the *number of collisions* of gas particles with each other and with the sides of the container they are in. So, if the volume decreases, the number of collisions increases and so the pressure will naturally increase. The opposite is true if the volume of the gas is increased. Now, the gas particles collide less frequently and the pressure will decrease.



Robert Boyle is the scientist who is credited with discovering that the pressure and volume of a sample of gas are **inversely proportional**. This can be seen when a graph of pressure against the inverse of volume is plotted. When the values are plotted, the graph is a straight line. This relationship is shown in Figure 7.4.

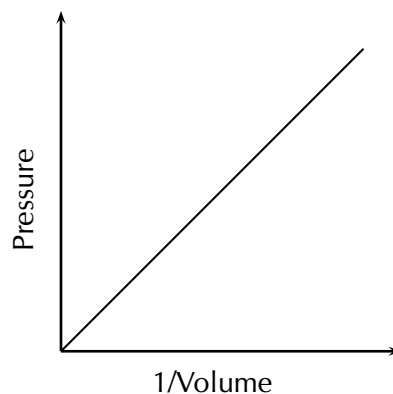


Figure 7.4: The graph of pressure plotted against the inverse of volume, produces a straight line. This shows that pressure and volume are inversely proportional.

We have just seen that the pressure of a gas is *inversely proportional* to the volume of the gas, provided the temperature stays the same. We can write this relationship symbolically as:

$$p \propto \frac{1}{V}$$

where $\propto$ means proportional and we write $\frac{1}{V}$ to show that the proportionality is inverse.

This equation can also be written as follows:

$$p = \frac{k}{V}$$

where k is a proportionality constant. If we rearrange this equation, we can say that:

$$pV = k$$

This equation means that, assuming the temperature and amount of gas is constant, multiplying any pressure and volume values for a fixed amount of gas will always give the same value (k). For example:

$$\begin{aligned} p_1 V_1 &= k \\ p_2 V_2 &= k \end{aligned}$$

where the subscripts 1 and 2 refer to two pairs of pressure and volume readings for the same mass of any gas at the same temperature.

From this, we can then say that:

$$\boxed{p_1 V_1 = p_2 V_2}$$

And we can also generalise this to say that:

$$\begin{aligned} p_1 V_1 &= p_2 V_2 \\ &= p_3 V_3 \\ &= p_n V_n \end{aligned}$$

In other words we can use any two pairs of readings, it does not have to be the first and second readings, it can be the first and third readings or the second and fifth readings.

If you work out the value of k for any pair of pressure readings from the experiment above and then work out k for any other pair of pressure readings you should find they are the same.

For example if you have:

$$p_1 = 1 \text{ atm}$$

$$p_2 = 2 \text{ atm}$$

$$V_1 = 4 \text{ cm}^3$$

$$V_2 = 2 \text{ cm}^3$$

then using the first pressure and volume gives:

$$k = p_1 V_1$$

$$k = (1 \text{ atm})(4 \text{ cm}^3)$$

$$= 4 \text{ atm} \cdot \text{cm}^3$$

and using the second pressure and volume gives:

$$k = p_2 V_2$$

$$= (2 \text{ atm})(2 \text{ cm}^3)$$

$$= 4 \text{ atm} \cdot \text{cm}^3$$

Remember that Boyle's Law requires two conditions. First, the **amount** of gas must stay constant. If you let a little of the air escape from the container in which it is enclosed, the pressure of the gas will decrease along with the volume, and the inverse proportion relationship is broken. Second, the **temperature** must stay constant. Cooling or heating matter generally causes it to contract or expand, or the pressure to decrease or increase. In the experiment, if we were to heat up the gas, it would expand and require you to apply a greater force to keep the plunger at a given position. Again, the proportionality would be broken.

▶ See simulation: 23VV at www.everythingscience.co.za

Before we look at some calculations using Boyle's law we first need to know some different units for volume and pressure. Volume units should be familiar to you from earlier grades and will usually be cm^3 or dm^3 or m^3 or L. The SI unit for volume is m^3 .

Pressure is measured in several different units. We can measure pressure in millimetres of mercury (mm Hg) or pascals (Pa) or atmospheres (atm). The SI unit for pressure is Pa. See Table 7.1 for converting between units.

FACT

Did you know that the mechanisms involved in *breathing* also relate to Boyle's Law? Just below the lungs is a muscle called the **diaphragm**. When a person breathes in, the diaphragm moves down and becomes more "flattened" so that the volume of the lungs can increase. When the lung volume *increases*, the pressure in the lungs *decreases* (Boyle's law). Since air always moves from areas of high pressure to areas of lower pressure, air will now be drawn into the lungs because the air pressure *outside* the body is higher than the pressure *in* the lungs. The opposite process happens when a person breathes out. Now, the diaphragm moves upwards and causes the volume of the lungs to *decrease*. The pressure in the lungs will *increase*, and the air that was in the lungs will be forced out towards the lower air pressure outside the body.

Worked example 1: Boyle's law

QUESTION

A sample of helium occupies a volume of 160 cm^3 at 100 kPa and 25°C . What volume will it occupy if the pressure is adjusted to 80 kPa and the temperature remains unchanged?

TIP

It is not necessary to convert to SI units for Boyle's law. Changing pressure and volume into different units involves *multiplication*. If you were to change the units in the above equation, this would involve multiplication on both sides of the equation, and so the conversions cancel each other out. However, although SI units don't have to be used, you must make sure that for each variable you use the *same* units throughout the equation. This is not true for some of the calculations we will do at a later stage, where SI units *must* be used.

SOLUTION

Step 1: Write down all the information that you know about the gas.

$$p_1 = 100 \text{ kPa}$$

$$p_2 = 80 \text{ kPa}$$

$$V_1 = 160 \text{ cm}^3$$

$$V_2 = ?$$

Step 2: Use an appropriate gas law equation to calculate the unknown variable.

Because the temperature of the gas stays the same, the following equation can be used:

$$p_2 V_2 = p_1 V_1$$

Step 3: Substitute the known values into the equation, making sure that the units for each variable are the *same*. Calculate the unknown variable.

$$(80)V_2 = (100)(160)$$

$$(80)V_2 = 16\,000$$

$$V_2 = 200 \text{ cm}^3$$

The volume occupied by the gas at a pressure of 80 kPa is 200 cm³

Step 4: Check your answer

Boyle's law states that the pressure is inversely proportional to the volume. Since the pressure decreased the volume must increase. Our answer for the volume is greater than the initial volume and so our answer is reasonable.

Worked example 2: Boyle's law**QUESTION**

The volume of a sample of gas is increased from 2,5 L to 2,8 L while a constant temperature is maintained. What is the final pressure of the gas under these volume conditions, if the initial pressure is 695 Pa?

SOLUTION

Step 1: Write down all the information that you know about the gas.

$$V_1 = 2,5 \text{ L}$$

$$V_2 = 2,8 \text{ L}$$

$$p_1 = 695 \text{ Pa}$$

$$p_2 = ?$$

Step 2: Choose a relevant gas law equation to calculate the unknown variable.

The sample of gas is at constant temperature and so we can use Boyle's law:

$$p_2 V_2 = p_1 V_1$$

Step 3: Substitute the known values into the equation, making sure that the units for each variable are the *same*. Calculate the unknown variable.

$$(2,8)p_2 = (695)(2,5)$$

$$(2,8)p_2 = 1737,5$$

$$p_2 = 620,5 \text{ kPa}$$

The pressure of the gas at a volume of 2,8 L is 620,5 kPa

Step 4: Check your answer

Boyle's law states that the pressure is inversely proportional to the volume. Since the volume increased the pressure must decrease. Our answer for the pressure is less than the initial pressure and so our answer is reasonable.

Exercise 7 – 2: Boyle's law

1. An unknown gas has an initial pressure of 150 kPa and a volume of 1 L. If the volume is increased to 1,5 L, what will the pressure be now?
2. A bicycle pump contains 250 cm³ of air at a pressure of 90 kPa. If the air is compressed, the volume is reduced to 200 cm³. What is the pressure of the air inside the pump?

3. The air inside a syringe occupies a volume of 10 cm^3 and exerts a pressure of 100 kPa. If the end of the syringe is sealed and the plunger is pushed down, the pressure increases to 120 kPa. What is the volume of the air in the syringe?
4. During an investigation to find the relationship between the pressure and volume of an enclosed gas at constant temperature, the following results were obtained.

Volume (dm^3)	12	16	20	24	28	32	36	40
Pressure (kPa)	400	300	240	200	171	150	133	120

- a) Plot a graph of pressure (p) against volume (V). Volume will be on the x-axis and pressure on the y-axis. Describe the relationship that you see.
- b) Plot a graph of p against $\frac{1}{V}$. Describe the relationship that you see.
- c) Do your results support Boyle's Law? Explain your answer.
5. Masoabi and Justine are experimenting with Boyle's law. They both used the same amount of gas. Their data is given in the table below:

	Masoabi		Justine	
	Initial	Final	Initial	Final
Temperature (K)	325	350	325	325
Volume (dm^3)	1	3	1	3
Pressure (Pa)	650	233	650	217

Masoabi and Justine argue about who is correct.

- a) Calculate the final pressure that would be expected using the initial pressure and volume and the final volume.
- b) Who correctly followed Boyle's law and why?

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1. [23VX](#) 2. [23VY](#) 3. [23VZ](#) 4. [23W2](#) 5. [23W3](#)



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Charles' law: Volume and temperature of an enclosed gas

ESBNX

Charles' law describes the relationship between the **volume** and **temperature** of a gas. The law was first published by Joseph Louis Gay-Lussac in 1802, but he referenced unpublished work by Jacques Charles from around 1787. This law states that at constant pressure, the volume of a given mass of an ideal gas increases or decreases by the same factor as its temperature (in Kelvin) increases or decreases. Another way of

saying this is that temperature and volume are **directly proportional**.

DEFINITION: *Charles' Law*

The volume of an enclosed sample of gas is directly proportional to its Kelvin temperature provided the pressure and amount of gas is kept constant.

Informal experiment: Charles's law

Aim:

To demonstrate Charles's Law.

Apparatus:

glass bottle (e.g. empty glass coke bottle), balloon, beaker or pot, water, hot plate

Method:

1. Place the balloon over the opening of the empty bottle.
2. Fill the beaker or pot with water and place it on the hot plate.
3. Stand the bottle in the beaker or pot and turn the hot plate on.
4. Observe what happens to the balloon.

Results:

You should see that the balloon starts to expand. As the air inside the bottle is heated, the pressure also increases, causing the volume to increase. Since the volume of the glass bottle can't increase, the air moves into the balloon, causing it to expand.

Conclusion:

The temperature and volume of the gas are directly related to each other. As one increases, so does the other.

You can also see this if you place the balloon and bottle into a freezer. The balloon will shrink after being in the freezer for a short while.

Mathematically, the relationship between temperature and pressure can be represented as follows:

$$V \propto T$$

In other words, the volume is directly proportional to the temperature.

Or, replacing the proportionality symbol with the constant of proportionality (k):

$$V = kT$$

If the equation is rearranged, then:

$$\frac{V}{T} = k$$

or:

$$\frac{V_1}{T_1} = k$$

$$\frac{V_2}{T_2} = k$$

$$\frac{V_n}{T_n} = k$$

So we can say that:

$$\boxed{\frac{V_1}{T_1} = \frac{V_2}{T_2}}$$

The equation relating volume and temperature produces a straight line graph.

However, if we plot this graph using the **Celsius** temperature scale (i.e. using °C), the zero point of temperature doesn't correspond to the zero point of volume. When the volume is zero, the temperature is actually $-273\text{ }^{\circ}\text{C}$ (Figure 7.5).

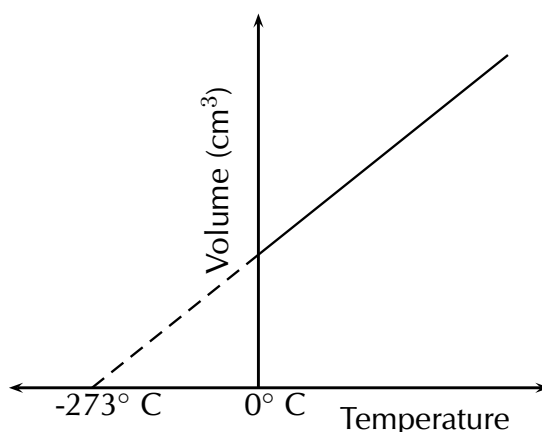


Figure 7.5: The relationship between volume and temperature, shown on the Celsius temperature scale.

A new temperature scale, the **Kelvin scale** must be used instead. Since zero on the Celsius scale corresponds with a Kelvin temperature of $-273\text{ }^{\circ}\text{C}$, it can be said that:

$$\text{Kelvin temperature } (T_K) = \text{Celsius temperature } (T_C) + 273$$

We can write:

$$T_K = T_C + 273 \quad \text{or} \quad T_C = T_K - 273$$

We can now plot the graph of temperature versus volume on the Kelvin scale. This is shown in Figure 7.6.

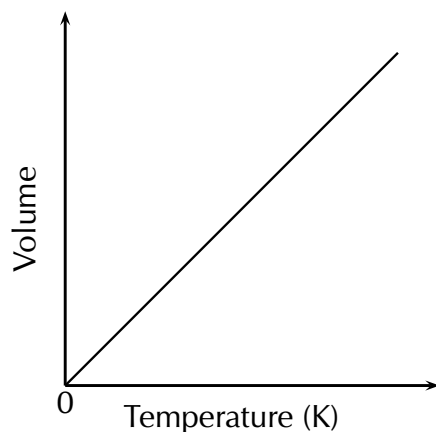


Figure 7.6: The volume of a gas is directly proportional to its temperature, provided the pressure of the gas is constant.

► See video: 23W4 at www.everythingscience.co.za

Can you explain Charles' law in terms of the kinetic theory of gases? When the temperature of a gas increases, so does the average speed of its molecules. The molecules collide with the walls of the container more often and with greater impact. These collisions will push back the walls, so that the gas occupies a greater volume than it did at the start. We saw this in the first demonstration. Because the glass bottle couldn't expand, the gas pushed out the balloon instead.

Worked example 3: Charles' law

QUESTION

At a temperature of 298 K, a certain amount of CO_2 gas occupies a volume of 6 L. What temperature will the gas be at if its volume is reduced to 5,5 L? The pressure remains constant.

SOLUTION

Step 1: Write down all the information that you know about the gas.

$$V_1 = 6 \text{ L}$$

$$V_2 = 5,5 \text{ L}$$

$$T_1 = 298 \text{ K}$$

$$T_2 = ?$$

Step 2: Convert the known values to SI units if necessary.

Temperature data is already in Kelvin, and so no conversions are necessary.

Step 3: Choose a relevant gas law equation that will allow you to calculate the unknown variable.

The pressure is kept constant while the volume and temperature are being varied. The amount of gas is also kept constant and so we can use Charles' law:

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Step 4: Substitute the known values into the equation. Calculate the unknown variable.

$$\begin{aligned}\frac{6}{298} &= \frac{5,5}{T_2} \\ 0,0201 \dots &= \frac{5,5}{T_2} \\ (0,0201 \dots)T_2 &= 5,5 \\ T_2 &= 273,2 \text{ K}\end{aligned}$$

The gas will be at a temperature of 273,2 K.

Step 5: Check your answer

Charles' law states that the temperature is directly proportional to the volume. In this example the volume decreases and so the temperature must decrease. Our answer gives a lower final temperature than the initial temperature and so is correct.

Worked example 4: Charles' law

QUESTION

Ammonium chloride and calcium hydroxide are allowed to react. The ammonia that is released in the reaction is collected in a gas syringe (a syringe that has very little friction so that the plunger can move freely) and sealed in. This gas is allowed to come to room temperature which is 20 °C. The volume of the ammonia is found to be 122 mL. It is now placed in a water bath set at 32 °C. What will be the volume reading after the syringe has been left in the bath for 1 hour (assume the plunger moves completely freely)? (By leaving the syringe for this length of time, we can be certain that the sample of gas is at the higher temperature.)

SOLUTION

Step 1: Write down all the information that you know about the gas.

TIP

Note that here the temperature must be converted to Kelvin since the change from degrees Celsius involves addition, not multiplication by a fixed conversion ratio (as is the case with pressure and volume.)

$$V_1 = 122 \text{ mL}$$

$$V_2 = ?$$

$$T_1 = 20 \text{ }^\circ\text{C}$$

$$T_2 = 7 \text{ }^\circ\text{C}$$

Step 2: Convert the known values to SI units if necessary.

Here, temperature must be converted into Kelvin, therefore:

$$T_1 = 20 + 273 = 293 \text{ K}$$

$$T_2 = 32 + 273 = 305 \text{ K}$$

Step 3: Choose a relevant gas law equation that will allow you to calculate the unknown variable.

The pressure is kept constant while the volume and temperature are being varied. The amount of gas is also kept constant and so we can use Charles' law:

$$\frac{V_2}{T_2} = \frac{V_1}{T_1}$$

Step 4: Substitute the known values into the equation. Calculate the unknown variable.

$$\begin{aligned}\frac{V_2}{305} &= \frac{122}{293} \\ \frac{V_2}{305} &= 0,416 \dots \\ V_2 &= 127 \text{ mL}\end{aligned}$$

The volume reading on the syringe will be 127 mL after the syringe has been left in the water bath for one hour.

Step 5: Check your answer

Charles' law states that the temperature is directly proportional to the volume. In this example the temperature increases and so the volume must increase. Our answer gives a higher final volume than the initial volume and so is correct.

TIP

You may see this law referred to as Gay-Lussac's law or as Amontons' law. Many scientists were working on the same problems at the same time and it is often difficult to know who actually discovered a particular law.

Exercise 7 – 3: Charles' law

1. The table below gives the temperature (in $^{\circ}\text{C}$) of helium gas under different volumes at a constant pressure.

Volume (L)	Temperature ($^{\circ}\text{C}$)
1,0	-161,9
1,5	-106,7
2	-50,8
2,5	4,8
3,0	60,3
3,5	115,9

- Draw a graph to show the relationship between temperature and volume.
 - Describe the relationship you observe.
 - If you extrapolate the graph (in other words, extend the graph line even though you may not have the exact data points), at what temperature does it intersect the x -axis?
 - What is significant about this temperature?
 - What conclusions can you draw? Use Charles' law to help you.
- A sample of nitrogen monoxide (NO) gas is at a temperature of 8°C and occupies a volume of $4,4\text{ dm}^3$. What volume will the sample of gas have if the temperature is increased to 25°C ?
 - A sample of oxygen (O_2) gas is at a temperature of 340 K and occupies a volume of $1,2\text{ dm}^3$. What temperature will the sample of gas be at if the volume is decreased to 200 cm^3 ?
 - Explain what would happen if you were verifying Charles' law and you let some of the gas escape.

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1. [23W5](#) 2. [23W6](#) 3. [23W7](#) 4. [23W8](#)



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Pressure-temperature relation

ESBNY

The pressure of a gas is directly proportional to its temperature, if the volume is kept constant (Figure 7.7). Recall that as the temperature of a gas increases, so does the kinetic energy of the particles in the gas. This causes the particles in the gas to move more rapidly and to collide with each other and with the side of the container more often. Since pressure is a measure of these collisions, the pressure of the gas increases with an increase in temperature. The pressure of the gas will decrease if its temperature decreases.

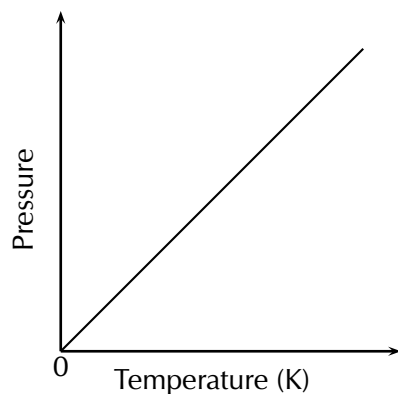


Figure 7.7: The relationship between the temperature and pressure of a gas.

In the same way that we have done for the other gas laws, we can describe the relationship between temperature and pressure using symbols, as follows:

$$T \propto p,$$

therefore:

$$p = kT$$

Rearranging this we get:

$$\frac{p}{T} = k$$

and that, provided the **amount** of gas stays the same (and the volume also stays the same):

$$\boxed{\frac{p_1}{T_1} = \frac{p_2}{T_2}}$$

Worked example 5: Pressure-temperature relation

QUESTION

At a temperature of 298 K, a certain amount of oxygen (O_2) gas has a pressure of 0,4 atm. What temperature will the gas be at if its pressure is increased to 0,7 atm?

SOLUTION

Step 1: Write down all the information that you know about the gas.

$$T_1 = 298 \text{ K}$$

$$T_2 = ?$$

$$p_1 = 0,4 \text{ atm}$$

$$p_2 = 0,7 \text{ atm}$$

Step 2: Convert the known values to SI units if necessary.

The temperature is already in Kelvin. We do not need to convert the pressure to pascals.

Step 3: Choose a relevant gas law equation that will allow you to calculate the unknown variable.

The volume is kept constant while the pressure and temperature are being varied. The amount of gas is also kept constant and so we can use the pressure-temperature relation:

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

Step 4: Substitute the known values into the equation. Calculate the unknown variable.

$$\begin{aligned}\frac{0,4}{298} &= \frac{0,7}{T_2} \\ 0,0013 \dots &= \frac{0,7}{T_2} \\ (0,0013 \dots)T_2 &= 0,7 \\ T_2 &= 521,5 \text{ K}\end{aligned}$$

The temperature will be 521,5 K.

Step 5: Check your answer

The pressure-temperature relation states that the pressure is directly proportional to the temperature. In this example the pressure increases and so the temperature must increase. Our answer gives a higher final temperature than the initial temperature and so is correct.

Worked example 6: Pressure-temperature relation

QUESTION

A fixed volume of carbon monoxide (CO) gas has a temperature of 32 °C and a pressure of 680 Pa. If the temperature is decreased to 15 °C what will the pressure be?

SOLUTION

Step 1: Write down all the information that you know about the gas.

$$\begin{aligned}
 T_1 &= 32\text{ }^{\circ}\text{C} \\
 T_2 &= 15\text{ }^{\circ}\text{C} \\
 p_1 &= 680\text{ Pa} \\
 p_2 &=?
 \end{aligned}$$

Step 2: Convert the known values to SI units if necessary.

We need to convert the given temperatures to Kelvin temperature.

$$\begin{aligned}
 T_1 &= 32 + 273 \\
 &= 305\text{ K} \\
 T_2 &= 15 + 273 \\
 &= 288\text{ K}
 \end{aligned}$$

Step 3: Choose a relevant gas law equation that will allow you to calculate the unknown variable.

$$\frac{p_2}{T_2} = \frac{p_1}{T_1}$$

Step 4: Substitute the known values into the equation. Calculate the unknown variable.

$$\begin{aligned}
 \frac{p_2}{288} &= \frac{680}{305} \\
 \frac{p_2}{288} &= 2,2295\dots \\
 p_2 &= 642,1\text{ Pa}
 \end{aligned}$$

The pressure will be 642,1 Pa.

Step 5: Check your answer

The pressure-temperature relation states that the pressure is directly proportional to the temperature. In this example the temperature decreases and so the pressure must decrease. Our answer gives a lower final pressure than the initial pressure and so is correct.

Exercise 7 – 4: Pressure-temperature relation

1. The table below gives the temperature (in $^{\circ}\text{C}$) of helium under different pressures

at a constant volume.

Pressure (atm)	Temperature (°C)	Temperature (K)
1,0	20	
1,2	78,6	
1,4	137,2	
1,6	195,8	
1,8	254,4	
2,0	313	

- Convert all the temperature data to Kelvin.
 - Draw a graph to show the relationship between temperature and pressure.
 - Describe the relationship you observe.
- A cylinder that contains methane gas is kept at a temperature of 15 °C and exerts a pressure of 7 atm. If the temperature of the cylinder increases to 25 °C, what pressure does the gas now exert?
 - A cylinder of propane gas at a temperature of 20 °C exerts a pressure of 8 atm. When a cylinder has been placed in sunlight, its temperature increases to 25 °C. What is the pressure of the gas inside the cylinder at this temperature?
 - A hairspray can is a can that contains a gas under high pressures. The can has the following warning written on it: "Do not place near open flame. Do not dispose of in a fire. Keep away from heat." Use what you know about the pressure and temperature of gases to explain why it is dangerous to not follow this warning.
 - A cylinder of acetylene gas is kept at a temperature of 291 K. The pressure in the cylinder is 5 atm. This cylinder can withstand a pressure of 8 atm before it explodes. What is the maximum temperature that the cylinder can safely be stored at?

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The general gas equation

ESBNZ

All the gas laws we have described so far rely on the fact that the amount of gas and one other variable (temperature, pressure or volume) remains constant. Since this is unlikely to be the case most times, it is useful to combine the relationships into one equation. We will use Boyle's law and the pressure-temperature relation to work out the general gas equation.

Boyle's law: $p \propto \frac{1}{V}$ (constant T)

In other words pressure is inversely proportional to the volume at constant temperature.

Pressure-temperature relation: $p \propto T$ (constant V)

In other words pressure is also directly proportional to the temperature at constant volume.

If we now vary both the volume and the temperature, the two proportionalities will still hold, but will be equal to a different proportionality constant.

So we can combine these relationships, to get:

$$p \propto \frac{T}{V}$$

Note that this says that the pressure is still directly proportional to the temperature and inversely proportional to the volume.

If we introduce the proportionality constant, k , we get:

$$p = k \frac{T}{V}$$

or, rearranging the equation:

$$pV = kT$$

We can also rewrite this relationship as follows:

$$\frac{pV}{T} = k$$

Provided the mass of the gas stays the same, we can also say that:

$$\boxed{\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}}$$

In the above equation, the subscripts 1 and 2 refer to two pressure and volume readings for the same mass of gas under different conditions. This is known as the **general gas equation**. Temperature is always in Kelvin and the units used for pressure and volume must be the same on both sides of the equation.

TIP

Remember that the general gas equation only applies if the mass (or number of moles) of the gas is fixed.

Worked example 7: General gas equation

QUESTION

At the beginning of a journey, a truck tyre has a volume of 30 dm^3 and an internal pressure of 170 kPa . The temperature of the tyre is 16°C . By the end of the trip, the volume of the tyre has increased to 32 dm^3 and the temperature of the air inside the tyre is 40°C . What is the tyre pressure at the end of the journey?

SOLUTION

Step 1: Write down all the information that you know about the gas.

$$\begin{aligned}
 T_1 &= 16\text{ }^{\circ}\text{C} \\
 T_2 &= 40\text{ }^{\circ}\text{C} \\
 V_1 &= 30\text{ dm}^3 \\
 V_2 &= 32\text{ dm}^3 \\
 p_1 &= 170\text{ kPa} \\
 p_2 &=?
 \end{aligned}$$

Step 2: Convert the known values to SI units if necessary.

We need to convert the given temperatures to Kelvin temperature.

$$\begin{aligned}
 T_1 &= 16 + 273 \\
 &= 289\text{ K} \\
 T_2 &= 40 + 273 \\
 &= 313\text{ K}
 \end{aligned}$$

Step 3: Choose a relevant gas law equation that will allow you to calculate the unknown variable.

Temperature, pressure and volume are all varying so we must use the general gas equation:

$$\frac{p_2 V_2}{T_2} = \frac{p_1 V_1}{T_1}$$

Step 4: Substitute the known values into the equation. Calculate the unknown variable.

$$\begin{aligned}
 \frac{(32)p_2}{313} &= \frac{(170)(30)}{289} \\
 \frac{(32)p_2}{313} &= 17,647\dots \\
 (32)p_2 &= 5523,529\dots \\
 p_2 &= 172,6\text{ kPa}
 \end{aligned}$$

The pressure will be 172,6 kPa.

QUESTION

A sample of a gas exerts a pressure of 100 kPa at 15 °C. The volume under these conditions is 10 dm³. The pressure increases to 130 kPa and the temperature increases to 32 °C. What is the new volume of the gas?

SOLUTION

Step 1: Write down all the information that you know about the gas.

$$T_1 = 15 \text{ °C}$$

$$T_2 = 32 \text{ °C}$$

$$p_1 = 100 \text{ kPa}$$

$$p_2 = 130 \text{ kPa}$$

$$V_1 = 10 \text{ dm}^3$$

$$V_2 = ?$$

Step 2: Convert the known values to SI units if necessary.

Here, temperature must be converted into Kelvin, therefore:

$$T_1 = 15 + 273 = 288 \text{ K}$$

$$T_2 = 32 + 273 = 305 \text{ K}$$

Step 3: Choose a relevant gas law equation that will allow you to calculate the unknown variable.

We use the general gas equation:

$$\frac{p_2 V_2}{T_2} = \frac{p_1 V_1}{T_1}$$

Step 4: Substitute the known values into the equation. Calculate the unknown variable.

$$\frac{(130)V_2}{305} = \frac{(100)(10)}{288}$$

$$\frac{(130)V_2}{305} = 3,47 \dots$$

$$(130)V_2 = 1059,027 \dots$$

$$V_2 = 8,15 \text{ dm}^3$$

The volume will be 8,15 dm³.

QUESTION

A cylinder of propane gas is kept at a temperature of 298 K. The gas exerts a pressure of 5 atm and the cylinder holds 4 dm³ of gas. If the pressure of the cylinder increases to 5,2 atm and 0,3 dm³ of gas leaks out, what temperature is the gas now at?



SOLUTION

Step 1: Write down all the information that you know about the gas.

$$T_1 = 298 \text{ K}$$

$$T_2 = ?$$

$$V_1 = 4 \text{ dm}^3$$

$$V_2 = 4 - 0,3 = 3,7 \text{ dm}^3$$

$$p_1 = 5 \text{ atm}$$

$$p_2 = 5,2 \text{ atm}$$

Step 2: Convert the known values to SI units if necessary.

Temperature data is already in Kelvin. All other values are in the same units.

Step 3: Choose a relevant gas law equation that will allow you to calculate the unknown variable.

Since volume, pressure and temperature are varying, we must use the general gas equation:

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$$

Step 4: Substitute the known values into the equation. Calculate the unknown variable.

$$\begin{aligned} \frac{(5)(4)}{298} &= \frac{(5,2)(3,7)}{T_2} \\ 0,067 \dots &= \frac{19,24}{T_2} \\ (0,067 \dots)T_2 &= 19,24 \\ T_2 &= 286,7 \text{ K} \end{aligned}$$

The temperature will be 286,7 K.

Exercise 7 – 5: The general gas equation

1. A closed gas system initially has a volume of 8 L and a temperature of 100 °C. The pressure of the gas is unknown. If the temperature of the gas decreases to 50 °C, the gas occupies a volume of 5 L and the pressure of the gas is 1,2 atm. What was the initial pressure of the gas?
2. A balloon is filled with helium gas at 27 °C and a pressure of 1,0 atm. As the balloon rises, the volume of the balloon increases by a factor of 1,6 and the temperature decreases to 15 °C. What is the final pressure of the gas (assuming none has escaped)?
3. 25 cm³ of gas at 1 atm has a temperature of 25 °C. When the gas is compressed to 20 cm³, the temperature of the gas increases to 28 °C. Calculate the final pressure of the gas.

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TIP

A joule can be defined as Pa·m³. So when you are using the ideal gas equation you must use the SI units to ensure that you get the correct answer.

The ideal gas equation

ESBP2

In the early 1800's, Amedeo Avogadro noted that if you have samples of different gases, of the same volume, at a fixed temperature and pressure, then the samples must contain the same number of freely moving particles (i.e. atoms or molecules).

DEFINITION: Avogadro's Law

Equal volumes of gases, at the same temperature and pressure, contain the same number of molecules.

You will remember from the previous section, that we combined different gas law equations to get *one* that included temperature, volume and pressure. In this equation

$$\frac{pV}{T} = k$$

the value of k is different for different masses of gas.

We find that when we calculate k for 1 mol of gas that we get:

$$\frac{pV}{T} = 8,314$$

This result is given a special name. It is the universal gas constant, R . R is measured in units of J·K⁻¹·mol⁻¹. No matter which gas we use, 1 mol of that gas will have the same constant.

TIP

All quantities in the equation $pV = nRT$ must be in the same units as the value of R . In other words, SI units must be used throughout the equation.

If we now extend this result to any number of moles of a gas we get the following:

$$\frac{pV}{T} = nR$$

where n is the number of moles of gas.

Rearranging this equation gives:

$$pV = nRT$$

This is the ideal gas equation. When you work with this equation you must have all units in SI units.

📺 See video: [23WK](https://www.everythingscience.co.za/23WK) at www.everythingscience.co.za

Worked example 10: Ideal gas equation

QUESTION

Two moles of oxygen (O_2) gas occupy a volume of 25 dm^3 at a temperature of 40°C . Calculate the pressure of the gas under these conditions.

SOLUTION

Step 1: Write down all the information that you know about the gas.

$$p = ?$$

$$V = 25 \text{ dm}^3$$

$$n = 2 \text{ mol}$$

$$T = 40^\circ\text{C}$$

$$R = 8,314 \text{ J}\cdot\text{K}\cdot\text{mol}^{-1}$$

Step 2: Convert the known values to SI units if necessary.

We need to convert the temperature to Kelvin and the volume to m^3 :

$$V = \frac{25}{1000} = 0,025 \text{ m}^3$$

$$T = 40 + 273 = 313 \text{ K}$$

Step 3: Choose a relevant gas law equation that will allow you to calculate the unknown variable.

We are varying everything (temperature, pressure, volume and amount of gas) and so we must use the ideal gas equation.

$$pV = nRT$$

Step 4: Substitute the known values into the equation. Calculate the unknown variable.

$$(0,025 \text{ m}^3)(p) = (2 \text{ mol})(8,314 \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1})(313 \text{ K})$$

$$(0,025 \text{ m}^3)(p) = 5204,564 \text{ Pa}\cdot\text{m}^3$$

$$p = 208\,182,56 \text{ Pa}$$

The pressure will be 208 182,56 Pa or 208,2 kPa.

Worked example 11: Ideal gas equation

QUESTION

Carbon dioxide (CO_2) gas is produced as a result of the reaction between calcium carbonate and hydrochloric acid. The gas that is produced is collected in a container of unknown volume. The pressure of the gas is 105 kPa at a temperature of 20 °C. If the number of moles of gas collected is 0,86 mol, what is the volume?

SOLUTION

Step 1: Write down all the information that you know about the gas.

$$p = 105 \text{ kPa}$$

$$V = ?$$

$$n = 0,86 \text{ mol}$$

$$T = 20 \text{ }^\circ\text{C}$$

$$R = 8,314 \text{ J}\cdot\text{K}\cdot\text{mol}^{-1}$$

Step 2: Convert the known values to SI units if necessary.

We need to convert the temperature to Kelvin and the pressure to Pa:

$$p = 105 \times 1000 = 105\,000 \text{ Pa}$$

$$T = 20 + 273 = 293 \text{ K}$$

Step 3: Choose a relevant gas law equation that will allow you to calculate the unknown variable.

We are varying everything (temperature, pressure, volume and amount of gas) and so we must use the ideal gas equation.

$$pV = nRT$$

Step 4: Substitute the known values into the equation. Calculate the unknown variable.

$$(105\,000\text{ Pa})V = (8,314\text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1})(293\text{ K})(0,86\text{ mol})$$

$$(105\,000\text{ Pa})V = 2094,96\text{ Pa}\cdot\text{m}^3$$

$$V = 0,020\text{ m}^3$$

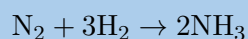
$$= 20\text{ dm}^3$$

The volume is 20 dm³.

Worked example 12: Ideal gas equation

QUESTION

Nitrogen (N₂) reacts with hydrogen (H₂) according to the following equation:



2 mol ammonia (NH₃) gas is collected in a separate gas cylinder which has a volume of 25 dm³. The pressure of the gas is 195,89 kPa. Calculate the temperature of the gas inside the cylinder.

SOLUTION

Step 1: Write down all the information that you know about the gas.

$$p = 195,89\text{ Pa}$$

$$V = 25\text{ dm}^3$$

$$n = 2\text{ mol}$$

$$R = 8,3\text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$$

$$T = ?$$

Step 2: Convert the known values to SI units if necessary.

We must convert the volume to m^3 and the pressure to Pa:

$$\begin{aligned}V &= \frac{25}{1000} \\&= 0,025 \text{ m}^3 \\p &= 195,89 \times 1000 \\&= 195\,890 \text{ Pa}\end{aligned}$$

Step 3: Choose a relevant gas law equation that will allow you to calculate the unknown variable.

$$pV = nRT$$

Step 4: Substitute the known values into the equation. Calculate the unknown variable.

$$\begin{aligned}(195\,890)(0,025) &= (2)(8,314)T \\4897,25 &= 16,628(T) \\T &= 294,52 \text{ K}\end{aligned}$$

The temperature is 294,52 K.

Worked example 13: Ideal gas equation

QUESTION

Calculate the number of moles of air particles in a classroom of length 10 m, a width of 7 m and a height of 2 m on a day when the temperature is 23°C and the air pressure is 98 kPa .

SOLUTION

Step 1: Calculate the volume of air in the classroom

The classroom is a rectangular prism (recall grade 10 maths on measurement). We can calculate the volume using:

$$\begin{aligned}V &= \text{length} \times \text{width} \times \text{height} \\&= (10)(7)(2) \\&= 140 \text{ m}^3\end{aligned}$$

Step 2: Write down all the information that you know about the gas.

$$p = 98 \text{ kPa}$$

$$V = 140 \text{ m}^3$$

$$n = ?$$

$$R = 8,314 \text{ J}\cdot\text{K}^{-1}\text{mol}^{-1}$$

$$T = 23 \text{ }^{\circ}\text{C}$$

Step 3: Convert the known values to SI units if necessary.

We must convert the temperature to K and the pressure to Pa:

$$T = 23 + 273 = 298 \text{ K}$$

$$p = 98 \times 1000 = 98\,000 \text{ Pa}$$

Step 4: Choose a relevant gas law equation that will allow you to calculate the unknown variable.

$$pV = nRT$$

Step 5: Substitute the known values into the equation. Calculate the unknown variable.

$$(98\,000)(140) = n(8,314)(298)$$

$$13\,720\,000 = 2477,572(n)$$

$$n = 5537,7 \text{ mol}$$

The number of moles in the classroom is 5537,7 mol.

Exercise 7 – 6: The ideal gas equation

1. An unknown gas has a pressure of 0,9 atm, a temperature of 120 °C and the number of moles is 0,28 mol. What is the volume of the sample?
2. 6 g of chlorine (Cl_2) occupies a volume of 0,002 m³ at a temperature of 26 °C. What is the pressure of the gas under these conditions?

3. An average pair of human lungs contains about 3,5 L of air after inhalation and about 3,0 L after exhalation. Assuming that air in your lungs is at 37 °C and 1,0 atm, determine the number of moles of air in a typical breath.

4. A learner is asked to calculate the answer to the problem below:

Calculate the pressure exerted by 1,5 moles of nitrogen gas in a container with a volume of 20 dm³ at a temperature of 37 °C.

The learner writes the solution as follows:

$$V = 20 \text{ dm}^3$$

$$n = 1,5 \text{ mol}$$

$$R = 8,314 \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$$

$$T = 37 + 273 = 310 \text{ K}$$

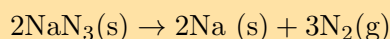
$$pT = nRV$$

$$p(310) = (1,5)(8,314)(20)$$

$$p(310) = (249,42)$$

$$= 0,8 \text{ kPa}$$

- a) Identify 2 mistakes the learner has made in the calculation.
b) Are the units of the final answer correct?
c) Rewrite the solution, correcting the mistakes to arrive at the right answer.
5. Most modern cars are equipped with airbags for both the driver and the passenger. An airbag will completely inflate in 0,05 s. This is important because a typical car collision lasts about 0,125 s. The following reaction of sodium azide (a compound found in airbags) is activated by an electrical signal:



- a) Calculate the mass of N₂(g) needed to inflate a sample airbag to a volume of 65 dm³ at 25 °C and 99,3 kPa. Assume the gas temperature remains constant during the reaction.
b) The above reaction produces heat, which raises the temperature in the airbag. Describe, in terms of the kinetic theory of gases, how the pressure in the sample airbag will change, if at all, as the gas temperature returns to 25 °C.

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- The **kinetic theory of gases** helps to explain the behaviour of gases under different conditions.
- The **kinetic theory of gases** states that gases are made up of constantly moving particles that have attractive forces between them.
- The **pressure** of a gas is a measure of the number of collisions of the gas particles with each other and with the sides of the container that they are in.
- The **temperature** of a substance is a measure of the average kinetic energy of the particles.
- An **ideal gas** has identical particles of zero volume, with no intermolecular forces between the particles. The atoms or molecules in an ideal gas move at the same speed.
- A **real gas** behaves like an ideal gas, except at high pressures and low temperatures. At low temperatures, the forces between molecules become significant and the gas will liquefy. At high pressures, the volume of the particles becomes significant.
- **Boyle's law** states that the pressure of a fixed quantity of gas is inversely proportional to the volume it occupies so long as the temperature remains constant. In other words, $pV = k$ or:

$$p_1 V_1 = p_2 V_2$$

- **Charles' law** states that the volume of an enclosed sample of gas is directly proportional to its Kelvin temperature provided the pressure and amount of gas remains constant. In other words, $\frac{V}{T} = k$ or:

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

- The **pressure** of a fixed mass of gas is directly proportional to its temperature, if the volume is constant. In other words, $\frac{p}{T} = k$ or:

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

- For Charles' law and for the pressure-temperature relation the temperature must be written in **Kelvin**. Temperature in degrees Celsius ($^{\circ}\text{C}$) can be converted to temperature in Kelvin (K) using the following equation:

$$T_K = T_C + 273$$

- Combining Boyle's law and the relationship between the temperature and pressure of a gas, gives the **general gas equation**, which applies as long as the amount of gas remains constant. The general gas equation is $\frac{pV}{T} = k$, or:

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$$

- **Avogadro's** law states that equal volumes of gases, at the same temperature and pressure, contain the same number of molecules.
- The **universal gas constant** (R) is $8,314 \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$. This constant is found by calculating $\frac{pV}{T}$ for 1 mol of any gas.
- Extending the above calculation to apply to any number of moles of gas gives the **ideal gas equation**:

$$pV = nRT$$

In this equation, **SI units** must be used. The SI unit for volume is m^3 , for pressure it is Pa and for temperature it is K.

Physical Quantities		
Quantity	Unit name	Unit symbol
Moles (n)	moles	mol
Pressure (p)	pascals	Pa
Temperature (T)	kelvin	K
Volume (V)	meters cubed	m^3

Exercise 7 – 7:

1. Give one word or term for each of the following definitions.
 - a) A gas is that has identical particles of zero volume, with no intermolecular forces between the particles.
 - b) The law that states that the volume of a gas is directly proportional to the temperature of the gas, provided that the pressure and the amount of the gas remain constant.
 - c) A measure of the average kinetic energy of gas particles.
2. Which one of the following properties of a fixed quantity of a gas must be kept constant during an investigation of Boyle's law?
 - a) density
 - b) pressure
 - c) temperature
 - d) volume

(IEB 2003 Paper 2)
3. Three containers of *equal volume* are filled with *equal masses* of helium, nitrogen and carbon dioxide gas respectively. The gases in the three containers are all at the same *temperature*. Which one of the following statements is correct regarding the pressure of the gases?
 - a) All three gases will be at the same pressure
 - b) The helium will be at the greatest pressure
 - c) The nitrogen will be at the greatest pressure

d) The carbon dioxide will be at the greatest pressure

(IEB 2004 Paper 2)

4. The ideal gas equation is given by $pV = nRT$. Which one of the following conditions is true according to Avogadro's hypothesis?

a	$p \propto \frac{1}{V}$	($T = \text{constant}$)
b	$V \propto T$	($p = \text{constant}$)
c	$V \propto n$	($p, T = \text{constant}$)
d	$p \propto T$	($n = \text{constant}$)

(DoE Exemplar paper 2, 2007)

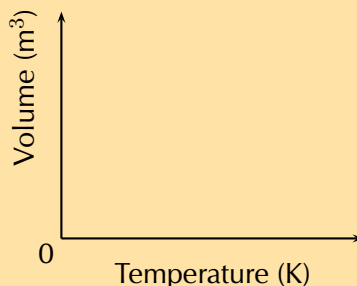
5. Complete the following table by stating whether or not the property is constant or variable for the given gas law.

Law	Pressure (p)	Volume (V)	Temperature (T)	Moles (n)
Boyle's law				
Charles' law				
Gay-Lussac's law				
General gas equation				
Ideal gas equation				

6. Use your knowledge of the gas laws to explain the following statements.

- It is dangerous to put an aerosol can near heat.
- A pressure vessel that is poorly designed and made can be a serious safety hazard (a pressure vessel is a closed, rigid container that is used to hold gases at a pressure that is higher than the normal air pressure).
- The volume of a car tyre increases after a trip on a hot road.

7. Copy the following set of labelled axes and answer the questions that follow:



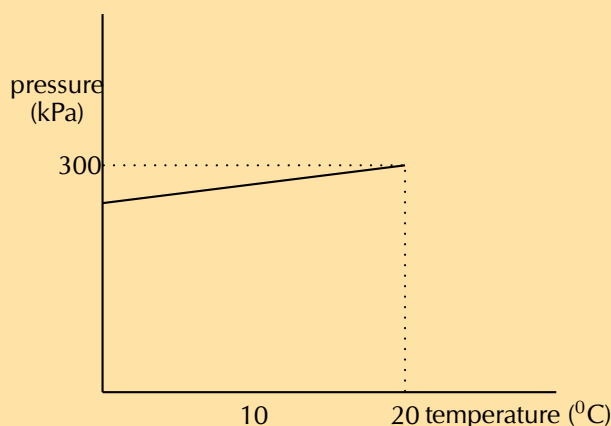
- On the axes, **using a solid line**, draw the graph that would be obtained for a fixed mass of an ideal gas if the pressure is kept constant.
- If the gradient of the above graph is measured to be $0,008 \text{ m}^3 \cdot \text{K}^{-1}$, calculate the pressure that 0,3 mol of this gas would exert.

(IEB 2002 Paper 2)

8. Two gas cylinders, A and B, have a volume of $0,15 \text{ m}^3$ and $0,20 \text{ m}^3$ respectively. Cylinder A contains 35 mol He gas at pressure p and cylinder B contains 40 mol He gas at 5 atm. The ratio of the Kelvin temperatures A:B is 1,80:1,00. Calculate the pressure of the gas (in kPa) in cylinder A.

(IEB 2002 Paper 2)

9. A learner investigates the relationship between the Celsius temperature and the pressure of a fixed amount of helium gas in a 500 cm^3 closed container. From the results of the investigation, she draws the graph below:



- Under the conditions of this investigation, helium gas behaves like an ideal gas. Explain briefly why this is so.
- From the shape of the graph, the learner concludes that the pressure of the helium gas is directly proportional to the Celsius temperature. Is her conclusion correct? Briefly explain your answer.
- Calculate the pressure of the helium gas at 0°C .
- Calculate the mass of helium gas in the container.

(IEB 2003 Paper 2)

10. One of the cylinders of a motor car engine, before compression contains 450 cm^3 of a mixture of air and petrol in the gaseous phase, at a temperature of 30°C and a pressure of 100 kPa. If the volume of the cylinder after compression decreases to one tenth of the original volume, and the temperature of the gas mixture rises to 140°C , calculate the pressure now exerted by the gas mixture.
11. A gas of unknown volume has a temperature of 14°C . When the temperature of the gas is increased to 100°C , the volume is found to be 5,5 L. What was the initial volume of the gas?
12. A gas has an initial volume of 2600 mL and a temperature of 350 K.
- If the volume is reduced to 1500 mL, what will the temperature of the gas be in Kelvin?
 - Has the temperature *increased* or *decreased*?
 - Explain this change, using the kinetic theory of gases.
13. In an experiment to determine the relationship between pressure and temperature of a fixed mass of gas, a group of learners obtained the following results:

Pressure (kPa)	101	120	130,5	138
Temperature ($^\circ\text{C}$)	0	50	80	100
Total gas volume (cm^3)	250	250	250	250

- a) Draw a straight-line graph of pressure (on the dependent, y-axis) versus temperature (on the independent, x-axis) on a piece of graph paper. Plot the points. Give your graph a suitable heading.
- b) *A straight-line graph passing through the origin is essential to obtain a mathematical relationship between pressure and temperature..*
Extrapolate (extend) your graph and determine the temperature (in °C) at which the graph will pass through the temperature axis.
- c) Write down, in words, the relationship between pressure and Kelvin temperature.
- d) From your graph, determine the pressure (in kPa) at 173 K. Indicate on your graph how you obtained this value.
- e) How would the gradient of the graph be affected (if at all) if a larger mass of the gas is used? Write down **ONLY increases, decreases or stays the same.**

(DoE Exemplar Paper 2, 2007)

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|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|-------------------------|
| 1a. 23WT | 1b. 23WV | 1c. 23WW | 2. 23WX | 3. 23WY | 4. 23WZ |
| 5. 23X2 | 6a. 23X3 | 6b. 23X4 | 6c. 23X5 | 7. 23X6 | 8. 23X7 |
| 9. 23X8 | 10. 23X9 | 11. 23XB | 12. 23XC | 13. 23XD | |



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Quantitative aspects of chemical change

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Wherever we look in real life we see the importance of mixing things in precise quantities. Cooking and baking, the medicines you take when sick, the products that you buy, all of these rely on the ingredients being mixed in specific amounts. And even the amount of product formed relies on how much of each ingredient is used. In this chapter we will look at some of these quantities and how they can be calculated.

In grade 10 we learnt about writing chemical equations and about the information that can be obtained from a balanced chemical equation. In this chapter we are going to explore these concepts further and learn more about gases, solutions and reactions. We will explore the concept of theoretical yield in greater detail and learn about limiting reagents.

Key Mathematics Concepts

- Ratio and proportion — Physical Sciences, Grade 10, Science skills
- Equations — Mathematics, Grade 10, Equations and inequalities
- Units and unit conversions — Physical Sciences, Grade 10, Science skills

8.1 Gases and solutions

ESBP4

We will begin by taking a closer look at gases and solutions and work out how to solve problems relating to them.

Molar volumes of gases

ESBP5

It is possible to calculate the volume of one mole of gas at standard temperature and pressure (STP) using what we now know about gases.

NOTE:

STP is a temperature of 273 K and a pressure of 101,3 kPa. The amount of gas is usually 1 mol.

We write down all the values that we know about one mole of gas at STP:

$$p = 101,3 \text{ kPa} = 101\,300 \text{ Pa}$$

$$n = 1 \text{ mol}$$

$$R = 8,31 \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$$

$$T = 273 \text{ K}$$

Now we can substitute these values into the ideal gas equation:

TIP

The standard units used for this equation are p in Pa, V in m³ and T in K. Remember also that 1000 cm³ = 1 dm³ and 1000 dm³ = 1 m³.

$$\begin{aligned}
 pV &= nRT \\
 (101\,300)V &= (1)(8,31)(273) \\
 (101\,300)V &= 2265,9 \\
 V &= 0,0224 \text{ m}^3 \\
 V &= 22,4 \text{ dm}^3
 \end{aligned}$$

The volume of 1 mole of gas at STP is 22,4 dm³.

And if we had any number of moles of gas, not just one mole then we would get:

$$V_g = 22,4n_g$$

Worked example 1: Molar gas volume

QUESTION

What is the volume of 2,3 mol of hydrogen gas at STP?

SOLUTION

Step 1: Find the volume

$$\begin{aligned}
 V_g &= (22,4)n_g \\
 &= (22,4)(2,3) \\
 &= 51,52 \text{ dm}^3
 \end{aligned}$$

Reactions and gases

ESBP6

Some reactions take place between gases. For these reactions we can work out the volumes of the gases using the fact that volume is proportional to the number of moles.

We can use the following formula:

$$V_A = \frac{a}{b} V_B$$

where:

V_A = volume of A

V_B = volume of B

a = stoichiometric coefficient of A

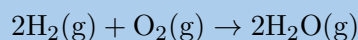
b = stoichiometric coefficient of B

TIP

The number in front of a reactant or a product in a balanced chemical equation is called the stoichiometric coefficient or stoichiometric ratio.

Worked example 2: Volume and gases**QUESTION**

Hydrogen and oxygen react to form water according to the following equation:



If 3 dm³ of oxygen is used, what volume of water is produced?

SOLUTION

Step 1: Determine the volume of water produced in the reaction.

We use the equation given above to work out the volume of water needed:

$$\begin{aligned} V_A &= \frac{a}{b} V_B \\ V_{\text{H}_2\text{O}} &= \frac{2}{1} V_{\text{O}_2} \\ &= 2(3) \\ &= 6 \text{ dm}^3 \end{aligned}$$

We can interpret the chemical equation in the worked example above ($2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g})$) as:

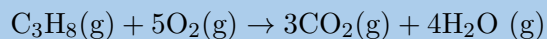
2 moles of hydrogen react with 1 mole of oxygen to produce 2 moles of water. We can also say that 2 volumes of hydrogen react with 1 volume of oxygen to produce 2 volumes of water.

Worked example 3: Gas phase calculations**QUESTION**

What volume of oxygen at STP is needed for the complete combustion of 3,3 dm³ of propane (C₃H₈)? (Hint: CO₂ and H₂O are the products as in all combustion reactions)

SOLUTION

Step 1: Write a balanced equation for the reaction.



Step 2: Determine the volume of oxygen needed for the reaction.

We use the equation given above to work out the volume of oxygen needed:

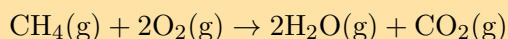
$$\begin{aligned}V_A &= \frac{a}{b}V_B \\V_{O_2} &= \frac{5}{1}V_{C_3H_8} \\&= 5(3,3) \\&= 16,5 \text{ dm}^3\end{aligned}$$

TIP

When you are busy with these calculations, you will need to remember the following:
 $1 \text{ dm}^3 = 1 \text{ L} = 1000 \text{ mL} = 1000 \text{ cm}^3$, therefore dividing a volume in cm^3 by 1000 will give you the equivalent volume in dm^3 .

Exercise 8 – 1:

1. Methane burns in oxygen, forming water and carbon dioxide according to the following equation:



If 4 dm^3 of methane is used, what volume of water is produced?

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Solutions

ESBP7

In grade 10 you learnt how to calculate the molar concentration of a solution. The molar concentration of a solution is the number of moles of solute per litre of solvent ($\text{mol}\cdot\text{L}^{-1}$). This is more commonly given as moles of solute per cubic decimetre of solution ($\text{mol}\cdot\text{dm}^{-3}$).

To calculate concentration we use $C = \frac{n}{V}$, where C is the molar concentration, n is the number of moles and V is the volume of the solution.

Calculating molar concentrations is useful to determine how much solute we need to add to a given volume of solvent in order to make a standard solution.

A standard solution is a solution in which the concentration is known to a high degree of precision. When we work with standard solutions we can take the concentration to be constant.

TIP

When performing a titration we say that the substance of unknown concentration is **titrated** with the standard solution. A pipette is a measuring device that is used to measure an exact amount of a liquid. If you use a pipette to add liquid to a flask then you would say that the liquid was **pipetted** into a flask.

Worked example 4: Concentration calculations**QUESTION**

How much sodium chloride (in g) will one need to prepare 500 cm³ of a standard solution with a concentration of 0,01 mol·dm⁻³?

SOLUTION

Step 1: Convert all quantities into the correct units for this equation.

The volume must be converted to dm³:

$$\begin{aligned} V &= \frac{500}{1000} \\ &= 0,5 \text{ dm}^3 \end{aligned}$$

Step 2: Calculate the number of moles of sodium chloride needed.

$$\begin{aligned} C &= \frac{n}{V} \\ 0,01 &= \frac{n}{0,5} \\ n &= 0,005 \text{ mol} \end{aligned}$$

Step 3: Convert moles of NaCl to mass.

To find the mass of NaCl we need the molar mass of NaCl. We can get this from the periodic table (recall from grade 10 how to calculate the molar mass of a compound).

$$\begin{aligned} m &= nM \\ &= (0,005)(58) \\ &= 0,29 \text{ g} \end{aligned}$$

The mass of sodium chloride needed is 0,29 g

We will now look at another use of concentration which is for titration calculations.

Titration

A titration is a technique for determining the concentration of an unknown solution. Titrations can be done using many different types of reactions. Acid-base reactions and redox reactions are both commonly used for titrations.

In grade 10 you did a simple acid-base titration. Now we will look at how to calculate the concentration of an unknown solution using an acid-base titration.

We can reduce the number of calculations that we have to do in titration calculations

by using the following:

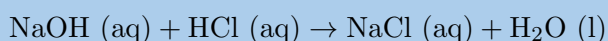
$$\frac{C_A V_A}{a} = \frac{C_B V_B}{b}$$

The a and b are the stoichiometric coefficients of compounds A and B respectively.

Worked example 5: Titration calculation

QUESTION

Given the equation:



25 cm³ of a 0,2 mol·dm⁻³ hydrochloric acid solution was pipetted into a conical flask and titrated with sodium hydroxide. It was found that 15 cm³ of the sodium hydroxide was needed to neutralise the acid. Calculate the concentration of the sodium hydroxide.

SOLUTION

Step 1: Write down all the information you know about the reaction, and make sure that the equation is balanced.

NaOH: $V = 15 \text{ cm}^3$

HCl: $V = 25 \text{ cm}^3$; $C = 0,2 \text{ mol}\cdot\text{dm}^{-3}$

The equation is already balanced.

Step 2: Convert the volume to dm³

$$\begin{aligned} V_{\text{NaOH}} &= \frac{15}{1000} \\ &= 0,015 \text{ dm}^3 \end{aligned}$$

$$\begin{aligned} V_{\text{HCl}} &= \frac{25}{1000} \\ &= 0,025 \text{ dm}^3 \end{aligned}$$

Step 3: Calculate the concentration of the sodium hydroxide

$$\begin{aligned} \frac{C_A V_A}{a} &= \frac{C_B V_B}{b} \\ \frac{(0,2)(0,025)}{1} &= \frac{(C_{\text{NaOH}})(0,015)}{1} \\ 0,005 &= (0,015)C_{\text{NaOH}} \\ C_{\text{NaOH}} &= 0,33 \text{ mol}\cdot\text{dm}^{-3} \end{aligned}$$

The concentration of the NaOH solution is 0,33 mol·dm⁻³

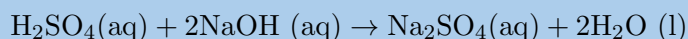
Worked example 6: Titration calculation

QUESTION

4,9 g of sulfuric acid is dissolved in water and the final solution has a volume of 220 cm³. Using an acid-base titration, it was found that 20 cm³ of this solution was able to completely neutralise 10 cm³ of a sodium hydroxide solution. Calculate the concentration of the sodium hydroxide in mol·dm⁻³.

SOLUTION

Step 1: Write a balanced equation for the titration reaction.



Step 2: Calculate the concentration of the sulfuric acid solution.

First convert the volume into dm³:

$$V = \frac{220}{1000} = 0,22 \text{ dm}^3$$

Then calculate the number of moles of sulfuric acid:

$$\begin{aligned} n &= \frac{m}{M} \\ &= \frac{4,9}{98} \\ &= 0,05 \text{ mol} \end{aligned}$$

Now we can calculate the concentration of the sulfuric acid:

$$\begin{aligned} C &= \frac{n}{V} \\ &= \frac{0,05}{0,22} \\ &= 0,227 \text{ mol} \cdot \text{dm}^{-3} \end{aligned}$$

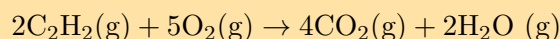
Step 3: Calculate the concentration of the sodium hydroxide solution.

Remember that only 20 cm³ or 0,02 dm³ of the sulfuric acid solution is used.

$$\begin{aligned} \frac{C_1 V_1}{n_1} &= \frac{C_2 V_2}{n_2} \\ \frac{(0,227)(0,02)}{1} &= \frac{(C_{\text{NaOH}})(0,01)}{2} \\ 0,00454 &= (0,005)C_{\text{NaOH}} \\ C_{\text{NaOH}} &= 0,909 \text{ mol} \cdot \text{dm}^{-3} \end{aligned}$$

Exercise 8 – 2: Gases and solutions

1. Acetylene (C_2H_2) burns in oxygen according to the following reaction:

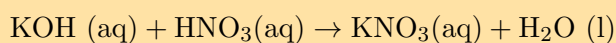


If $3,5 \text{ dm}^3$ of acetylene gas is burnt, what volume of carbon dioxide will be produced?

2. 130 g of magnesium chloride (MgCl_2) is dissolved in 300 mL of water.

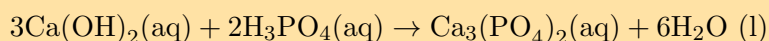
- Calculate the concentration of the solution.
- What mass of magnesium chloride would need to be added for the concentration to become $6,7 \text{ mol}\cdot\text{dm}^{-3}$?

3. Given the equation:



20 cm^3 of a $1,3 \text{ mol}\cdot\text{dm}^{-3}$ potassium hydroxide (KOH) solution was pipetted into a conical flask and titrated with nitric acid (HNO_3). It was found that 17 cm^3 of the nitric acid was needed to neutralise the base. Calculate the concentration of the nitric acid.

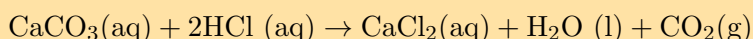
4. Given the equation:



10 cm^3 of a $0,4 \text{ mol}\cdot\text{dm}^{-3}$ calcium hydroxide ($\text{Ca}(\text{OH})_2$) solution was pipetted into a conical flask and titrated with phosphoric acid (H_3PO_4). It was found that 11 cm^3 of the phosphoric acid was needed to neutralise the base. Calculate the concentration of the phosphoric acid.

5. A $3,7 \text{ g}$ sample of an antacid (which contains mostly calcium carbonate) is dissolved in water. The final solution has a volume of 500 mL. 25 mL of this solution is then pipetted into a conical flask and titrated with hydrochloric acid. It is found that 20 mL of the hydrochloric acid completely neutralises the antacid solution. What is the concentration of the hydrochloric acid?

The equation for this reaction is:



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In grade 10 you learnt how to write balanced chemical equations and started looking at stoichiometric calculations. By knowing the ratios of substances in a reaction, it is possible to use stoichiometry to calculate the amount of either reactants or products that are involved in the reaction.

The following figure highlights the relation between the balanced chemical equation and the number of moles:

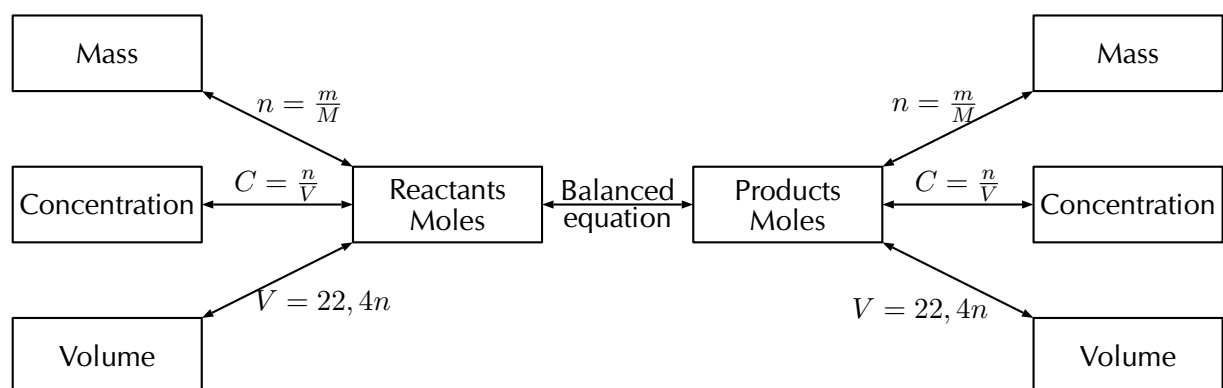


Figure 8.1: Stoichiometric flow diagram

In grade 10 we explored some of the concepts of stoichiometry. We looked at how to calculate the number of moles of a substance and how to find the molar mass. We also looked at how to find the molecular and empirical formulae of substances. Now we will explore more of these concepts such as limiting reagents, percent purity and percent yield.

Limiting reagents

ESBP9

Activity: What is a limiting reagent?

For this activity you will need A4 sheets of paper in white, red, blue, yellow, green and pink. (or you can use several sheets of white paper and colour them using kokis or crayons).

Tear the white sheet into five pieces, the red sheet into ten pieces, the blue sheet into eight pieces, the yellow sheet into seven pieces, the green sheet into nine pieces and the pink piece into four pieces.

1. Stick two red pieces to each white piece. Do you have any red or white pieces left?
2. Stick one yellow piece to each blue piece. Do you have any yellow or blue pieces left?

3. Stick three green pieces to each pink piece. Do you have any green or pink pieces left?

You should find that you had no red or white pieces left. For the blue and yellow pieces you should have one blue piece left. And for the green and pink pieces you should have had one pink piece left.

We say that the pink and blue pieces were in excess while the green and yellow sheets were limiting. In other words you would have had to tear the green and yellow sheets into **more** pieces or you would have had to tear the blue and pink pieces into **less** pieces.

In the above activity we could solve the problem of having too many or too few pieces of paper by simply tearing the pieces of paper into more pieces. In chemistry we also encounter this problem when mixing different substances. Often we will find that we added too much or too little of a particular substance. It is important to know that this happens and to know how much (i.e. the quantities) of different reactants are used in the reaction. This knowledge is used in industrial reactions.

DEFINITION: *Limiting reagent*

A limiting reagent (or reactant) is a reagent that is completely used up in a chemical reaction.

DEFINITION: *Excess reagent*

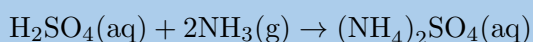
An excess reagent (or reactant) is a reagent that is not completely used up in a chemical reaction.

▶ See simulation: 23XN at www.everythingscience.co.za

Worked example 7: Limiting reagents

QUESTION

Sulfuric acid (H_2SO_4) reacts with ammonia (NH_3) to produce the fertiliser ammonium sulfate ($(\text{NH}_4)_2\text{SO}_4$) according to the following equation:



What is the maximum mass of ammonium sulfate that can be obtained from 2,0 kg of sulfuric acid and 1,0 kg of ammonia?

SOLUTION

Step 1: Convert the mass of sulfuric acid and ammonia into moles

Moles of sulfuric acid:

$$\begin{aligned}n &= \frac{m}{M} \\&= \frac{2000}{98} \\&= 20,4 \text{ mol}\end{aligned}$$

Moles of ammonia:

$$\begin{aligned}n &= \frac{m}{M} \\&= \frac{1000}{17} \\&= 58,8 \text{ mol}\end{aligned}$$

Step 2: Use the balanced equation to determine which of the reactants is limiting.

We need to look at how many moles of product we can get from each reactant. Then we compare these two results. The smaller number is the amount of product that we can produce and the reactant that gives the smaller number, is the limiting reagent.

The mole ratio of H_2SO_4 to $(\text{NH}_4)_2\text{SO}_4$ is 1 : 1. So the number of moles of $(\text{NH}_4)_2\text{SO}_4$ that can be produced from the sulfuric acid is:

$$\begin{aligned}n_{(\text{NH}_4)_2\text{SO}_4} &= n_{\text{H}_2\text{SO}_4} \times \frac{\text{stoichiometric coefficient } (\text{NH}_4)_2\text{SO}_4}{\text{stoichiometric coefficient } \text{H}_2\text{SO}_4} \\&= 20,4 \text{ mol } \text{H}_2\text{SO}_4 \times \frac{1 \text{ mol } (\text{NH}_4)_2\text{SO}_4}{1 \text{ mol } \text{H}_2\text{SO}_4} \\&= 20,4 \text{ mol } (\text{NH}_4)_2\text{SO}_4\end{aligned}$$

The mole ratio of NH_3 to $(\text{NH}_4)_2\text{SO}_4$ is 2 : 1. So the number of moles of $(\text{NH}_4)_2\text{SO}_4$ that can be produced from the ammonia is:

$$\begin{aligned}n_{(\text{NH}_4)_2\text{SO}_4} &= n_{\text{NH}_3} \times \frac{\text{stoichiometric coefficient } (\text{NH}_4)_2\text{SO}_4}{\text{stoichiometric coefficient } \text{NH}_3} \\&= 58,8 \text{ mol } \text{NH}_3 \times \frac{1 \text{ mol } (\text{NH}_4)_2\text{SO}_4}{2 \text{ mol } \text{NH}_3} \\&= 29,4 \text{ mol } (\text{NH}_4)_2\text{SO}_4\end{aligned}$$

Since we get less $(\text{NH}_4)_2\text{SO}_4$ from H_2SO_4 than is produced from NH_3 , the sulfuric acid is the limiting reactant.

Step 3: Calculate the maximum mass of ammonium sulfate that can be produced

From the step above we saw that we have 20,4 mol of $(\text{NH}_4)_2\text{SO}_4$.

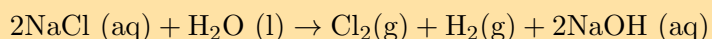
The maximum mass of ammonium sulfate that can be produced is calculated as follows:

$$\begin{aligned}m &= nM \\&= (20,4)(132) \\&= 2692,8 \text{ g} \\&= 2,6928 \text{ kg}\end{aligned}$$

The maximum mass of ammonium sulfate that can be produced is 2,69 kg.

Exercise 8 – 3:

1. When an electrical current is passed through a sodium chloride solution, sodium hydroxide can be produced according to the following equation:



What is the maximum mass of sodium hydroxide that can be obtained from 4,0 kg of sodium chloride and 3,0 kg of water?

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Percent yield

ESBPB

The percent yield of a reaction is very important as it tells us how efficient a reaction is. A reaction that has a low percent yield is not very useful in industry. If you are making a new medicine or pesticide and your reaction has a low percent yield then you would search for a different way of doing the reaction. This reduces the amount of (often very expensive) chemicals that you use and reduces waste.

The percent yield can be calculated using:

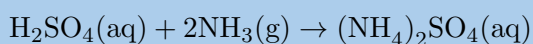
$$\% \text{yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

where the actual yield is the amount of product that is produced when you carry out the reaction and the theoretical yield is the amount of product that you calculate for the reaction using stoichiometric methods.

Worked example 8: Percent yield

QUESTION

Sulfuric acid (H_2SO_4) reacts with ammonia (NH_3) to produce the fertiliser ammonium sulfate ($(\text{NH}_4)_2\text{SO}_4$) according to the following equation:



A factory worker carries out the above reaction (using 2,0 kg of sulfuric acid and 1,0 kg of ammonia) and gets 2,5 kg of ammonium sulfate. What is the percentage yield of the reaction?

SOLUTION

Step 1: Determine which is the limiting reagent

We determined the limiting reagent for this reaction with the same amounts of reactants in the previous worked example, so we will just use the result from there.

Sulfuric acid is the limiting reagent. The number of moles of ammonium sulfate that can be produced is 20,4 mol.

Step 2: Calculate the theoretical yield of ammonium sulphate

From the previous worked example we found the maximum mass of ammonium sulfate that could be produced.

The theoretical yield (or maximum mass) of ammonium sulfate that can be produced is 2,69 kg.

Step 3: Calculate the percentage yield

$$\begin{aligned}\% \text{yield} &= \frac{\text{actual yield}}{\text{theoretical yield}} \times 100 \\ &= \frac{2,5}{2,694}(100) \\ &= 92,8\%\end{aligned}$$

This reaction has a high percent yield and so would therefore be a useful reaction to use in industry.

Exercise 8 – 4:

1. When an electrical current is passed through a sodium chloride solution, sodium hydroxide can be produced according to the following equation:



A chemist carries out the above reaction using 4,0 kg of sodium chloride and 3,0 kg of water. The chemist finds that they get 1,8 kg of sodium hydroxide. What is the percentage yield?

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Molecular and empirical formulae were introduced in grade 10. The empirical formula is the simplest formula of a compound (and represents the ratio of atoms of each element in a compound). The molecular formula is the full formula of the compound (and represents the total number of atoms of each element in a compound). You should also recall from grade 10 the percent composition of a substance. This is the percentage by molecular mass that each element contributes to the overall formula. For example water (H_2O) has the following percentage composition: 89% oxygen and 11% hydrogen.

Worked example 9: Empirical and molecular formula

QUESTION

Vinegar, which is used in our homes, is a dilute form of acetic acid. A sample of acetic acid has the following percentage composition: 39,9% carbon, 6,7% hydrogen and 53,4% oxygen.

1. Determine the empirical formula of acetic acid.
2. Determine the molecular formula of acetic acid if the molar mass of acetic acid is $60,06 \text{ g}\cdot\text{mol}^{-1}$.

SOLUTION

Step 1: Find the mass

In 100 g of acetic acid, there is: 39,9 g C, 6,7 g H and 53,4 g O.

Step 2: Find the moles

$$n = \frac{m}{M}$$

$$\begin{aligned}n_{\text{C}} &= \frac{39,9}{12} = 3,325 \text{ mol} \\n_{\text{H}} &= \frac{6,7}{1,01} = 6,6337 \text{ mol} \\n_{\text{O}} &= \frac{53,4}{16} = 3,3375 \text{ mol}\end{aligned}$$

Step 3: Find the empirical formula

To find the empirical formula we first note how many moles of each element we have. Then we divide the moles of each element by the smallest of these numbers, to get the ratios of the elements. This ratio is rounded off to the nearest whole number.

C	H	O
3,325	6,6337	3,3375
$\frac{3,325}{3,325}$	$\frac{6,6337}{3,325}$	$\frac{3,3375}{3,325}$
1	2	1

The empirical formula is CH₂O.

Step 4: Find the molecular formula

The molar mass of acetic acid using the empirical formula (CH₂O) is 30,02 g·mol⁻¹. However the question gives the molar mass as 60,06 g·mol⁻¹. Therefore the actual number of moles of each element must be double what it is in the empirical formula ($\frac{60,06}{30,02} = 2$). The molecular formula is therefore C₂H₄O₂ or CH₃COOH

Exercise 8 – 5:

1. A sample of oxalic acid has the following percentage composition: 26,7% carbon, 2,2% hydrogen and 71,1% oxygen.

Determine the molecular formula of oxalic acid if the molar mass of oxalic acid is 90 g·mol⁻¹.

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1. [23XR](#)



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Percent purity

ESBPD

The final use of stoichiometric calculations that we will look at is to determine the percent purity of a sample. Percent purity is important since when you make a compound you may have a small amount of impurity in the sample and you would need to keep this below a certain level. Or you may need to know how much of a particular ion is dissolved in water to determine if it is below the legally allowed level.

Percent purity can be calculated using:

$$\% \text{purity} = \frac{\text{mass of compound}}{\text{mass of sample}} \times 100$$

Worked example 10: Percent purity

QUESTION

Shells contain calcium carbonate (CaCO_3) as well as other minerals. Faarah wants to know how much calcium carbonate is in a shell. She finds that the shell weighs 5 g. After performing some more experiments she finds that the mass of calcium carbonate and the crucible (a container that is used to heat compounds in) is 3,2 g. The mass of the crucible is 0,5 g. How much calcium carbonate is in the shell?

SOLUTION

Step 1: Write down an equation for percent purity

Percent purity is given by:

$$\% \text{purity} = \frac{\text{mass of compound}}{\text{mass of sample}} \times 100$$

Step 2: Find the mass of the product

We are given the mass of the crucible and the mass of the crucible with the product. We need to subtract the mass of the crucible from the mass of the crucible with the product to obtain only the mass of the product.

$$\begin{aligned} \text{Mass product} &= 3,2 \text{ g} - 0,5 \text{ g} \\ &= 2,7 \text{ g} \end{aligned}$$

Step 3: Calculate the answer.

Substituting the calculated mass into the equation for percent purity gives:

$$\begin{aligned} \% \text{purity} &= \frac{\text{mass of compound}}{\text{mass of sample}} \times 100 \\ &= \frac{2,7}{5} (100) \\ &= 54\% \end{aligned}$$

Worked example 11: Percent purity

QUESTION

Limestone is mostly calcium carbonate (CaCO_3). Jake wants to know how much calcium carbonate is in a sample of limestone. He finds that the sample weighs 3,5 g. He then adds concentrated hydrochloric acid (HCl) to the sample. The equation for this

reaction is:



If the mass of calcium chloride produced is 3,6 g, what is the percent purity of the limestone sample?

SOLUTION

Step 1: Calculate the number of moles of calcium chloride

The number of moles of calcium chloride is:

$$\begin{aligned} n &= \frac{m}{M} \\ &= \frac{3,6}{111} \\ &= 0,032 \text{ mol} \end{aligned}$$

Step 2: Calculate the number of moles of calcium carbonate

The molar ratio of calcium chloride to calcium carbonate is 1:1. Therefore the number of moles of calcium carbonate is 0,032 mol.

Step 3: Calculate the mass of calcium carbonate

The mass of calcium carbonate is:

$$\begin{aligned} m &= nM \\ &= (0,032)(100) \\ m &= 3,24 \text{ g} \end{aligned}$$

Step 4: Calculate the percent purity

Substituting the calculated mass into the equation for percent purity gives:

$$\begin{aligned} \% \text{purity} &= \frac{\text{mass of compound}}{\text{mass of sample}} \times 100 \\ &= \frac{3,3}{3,5} \times (100) \\ &= 94,3\% \end{aligned}$$

Exercise 8 – 6:

1. Hematite contains iron oxide (Fe_2O_3) as well as other compounds. Thembile

wants to know how much iron oxide is in a sample of hematite. He finds that the sample of hematite weighs 6,2 g. After performing some experiments he finds that the mass of iron oxide and the crucible (a container that is used to heat compounds in) is 4,8 g. The mass of the crucible is 0,5 g. How much iron oxide is in the sample of hematite?

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1. [23XS](#)



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▶ See video: [23XT](#) at www.everythingscience.co.za

General experiment: The thermal decomposition of lead(II) nitrate

Aim:

To observe what happens when lead(II) nitrate is heated.

Apparatus:

- Bunsen burner
- Test tube (glass)
- Test tube holder
- 0,5 g lead(II) nitrate

Method:

WARNING!

Lead(II) nitrate produces toxic nitrogen dioxide on heating.

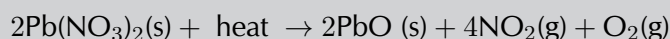
1. Place the lead(II) nitrate sample in the test tube.
2. Light the Bunsen burner and carefully hold the test tube in the flame. Remember to point the mouth of the test tube away from you.
3. Observe what happens.

Results:

A crackling noise is heard on heating the sample and a small amount of a brownish coloured gas is noted. The white powder became yellow.

If a glowing splint is held at the mouth of the test tube the splint reignites.

The balanced chemical for this reaction is:



Informal experiment:Stoichiometry

Aim:

To determine the percentage yield of magnesium carbonate from magnesium sulfate and sodium carbonate.

Apparatus:

- magnesium sulfate (Epsom salts)
- sodium carbonate
- mass meter
- hot plate
- small glass beakers (heat resistant)
- funnel
- filter paper

Method:

1. Weigh out 5 g of magnesium sulfate.
2. Dissolve the magnesium sulfate in 20 ml of water. Heat the solution until all the solid dissolves.
3. Dissolve 5 g of sodium carbonate in 20 ml of warm water. If necessary heat the solution to dissolve all the sodium carbonate.
4. Carefully pour the sodium carbonate solution into the hot magnesium sulfate solution. You should have a milky looking solution.
5. Weigh the filter paper (if your mass meter is not very accurate then assume the mass of the filter paper is 0). Carefully filter the final solution.
6. Leave the filter paper to dry slightly (or overnight) and then weigh it. The solid that stays on the filter paper is magnesium carbonate.

Results:

The equation for this reaction is: $\text{MgSO}_4(\text{aq}) + \text{Na}_2\text{CO}_3(\text{aq}) \rightarrow \text{Na}_2\text{SO}_4(\text{aq}) + \text{MgCO}_3(\text{s}) + \text{H}_2\text{O}(\text{l})$.

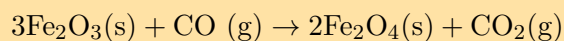
Use the above equation to work out the percentage yield of the magnesium sulfate. Remember that you need to determine which of the reactants is limiting. (You may get a percentage yield of greater than 100% if your sample is not completely dry.)

Conclusion:

By performing an experiment we were able to calculate the percentage yield of a reaction.

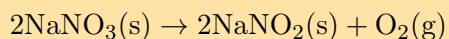
Exercise 8 – 7: Stoichiometry

1. Given the following reaction:



If 2,3 kg of Fe_2O_3 and 1,7 kg of CO is used, what is the maximum mass of Fe_2O_4 that can be produced?

2. Sodium nitrate decomposes on heating to produce sodium nitrite and oxygen according to the following equation:



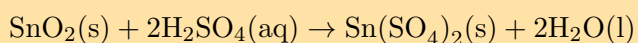
Nombusa carries out the above reaction using 50 g of sodium nitrate. Nombusa finds that they get 36 g of sodium nitrite. What is the percentage yield?

3. Benzene has the following percentage composition: 92,31% carbon and 7,69% hydrogen

Determine the molecular formula of benzene if the molar mass of benzene is $78 \text{ g}\cdot\text{mol}^{-1}$.

4. Cuprite is a minor ore of copper. Cuprite is mainly composed of copper(I) oxide (Cu_2O). Jennifer wants to know how much copper oxide is in a sample of cuprite. She has a sample of cuprite that weighs 7,7 g. She performs some experiments and finds that the mass of iron oxide and crucible (a container that is used to heat compounds in) is 7,4 g. The mass of the crucible is 0,2 g. What is the percent purity of the sample of cuprite?

5. A sample containing tin dioxide (SnO_2) is to be tested to see how much tin dioxide it contains. The sample weighs 6,2 g. Sulfuric acid (H_2SO_4) is added to the sample and tin sulfate ($\text{Sn}(\text{SO}_4)_2$) forms. The equation for this reaction is:



If the mass of tin sulfate produced is 4,7 g, what is the percent purity of the sample?

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1. 23XV 2. 23XW 3. 23XX 4. 23XY 5. 23XZ



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8.3 Volume relationships in gaseous reactions

ESBPF

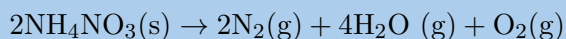
Using what we have learnt about stoichiometry and about gases we can now apply these principles to reactions involving gases.

We will use explosions as an example.

Worked example 12: Explosions

QUESTION

Ammonium nitrate is used as an explosive in mining. The following reaction occurs when ammonium nitrate is heated:



If 750 g of ammonium nitrate is used, what volume of oxygen gas would we expect to produce (at STP)?

SOLUTION

Step 1: Work out the number of moles of ammonium nitrate

The number of moles of ammonium nitrate used is:

$$\begin{aligned} n &= \frac{m}{M} \\ &= \frac{750}{80} \\ &= 9,375 \text{ mol} \end{aligned}$$

Step 2: Work out the amount of oxygen

The mole ratio of NH_4NO_3 to O_2 is 2 : 1. So the number of moles of O_2 is:

$$\begin{aligned} n_{\text{O}_2} &= n_{\text{NH}_4\text{NO}_3} \times \frac{\text{stoichiometric coefficient O}_2}{\text{stoichiometric coefficient NH}_4\text{NO}_3} \\ &= 9,375 \text{ mol NH}_4\text{NO}_3 \times \frac{1 \text{ mol O}_2}{2 \text{ mol NH}_4\text{NO}_3} \\ &= 4,6875 \text{ mol} \end{aligned}$$

Step 3: Work out the volume of oxygen

Recall from earlier in the chapter that we said that one mole of any gas occupies 22,4 dm³ at STP.

$$\begin{aligned}V &= (22,4)n \\&= (22,4)(4,6875) \\&= 105 \text{ dm}^3\end{aligned}$$

Airbags in cars use a controlled explosion to inflate the bag. When a car hits another car or an object, various sensors trigger the airbag. A chemical reaction then produces a large volume of gas which inflates the airbag.

Worked example 13: Controlled explosion

QUESTION

Sodium azide is sometimes used in airbags. When triggered, it has the following reaction:



If 55 grams of sodium azide is used, what volume of nitrogen gas would we expect to produce?

SOLUTION

Step 1: Work out the number of moles of sodium azide

The number of moles of sodium azide used is:

$$\begin{aligned}n &= \frac{m}{M} \\&= \frac{55}{65} \\&= 0,85 \text{ mol}\end{aligned}$$

Step 2: Work out the amount of nitrogen

The mole ratio of NaN_3 to N_2 is 2 : 3. So the number of moles of N_2 is:

$$\begin{aligned}n_{\text{N}_2} &= n_{\text{NaN}_3} \times \frac{\text{stoichiometric coefficient N}_2}{\text{stoichiometric coefficient NaN}_3} \\&= 0,85 \text{ mol NaN}_3 \times \frac{3 \text{ mol N}_2}{2 \text{ mol NaN}_3} \\&= 1,27 \text{ mol N}_2\end{aligned}$$

Step 3: Work out the volume of nitrogen

$$\begin{aligned}V &= (22,4)n \\&= (22,4)(1,27) \\&= 28,4 \text{ dm}^3\end{aligned}$$

Exercise 8 – 8: Gases 2

1. What volume of oxygen is needed for the complete combustion of 5 g of magnesium to form magnesium oxide?
2. Annalize is making a mini volcano for her science project. She mixes baking soda (mostly NaHCO_3) and vinegar (mostly CH_3COOH) together to make her volcano erupt. The reaction for this equation is:



What volume of carbon dioxide is produced if Annalize uses 50 ml of $0,2 \text{ mol}\cdot\text{dm}^3$ acetic acid?

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1. [23Y2](#) 2. [23Y3](#)



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8.4 Chapter summary

ESBPG

▶ See presentation: [23Y4](#) at www.everythingscience.co.za

- The volume of **one mole** of gas at STP is $22,4 \text{ dm}^3$

- For any number of moles of gas at STP we can use $V_g = 22,4n_g$ to find the volume.
- The volume relationship for two gases in a reaction is given by: $V_A = \frac{a}{b}V_B$.
where V_A is the volume of gas A, V_B is the gas B, a is the stoichiometric coefficient of gas A and b is the stoichiometric coefficient of gas B.
- The **concentration** of a solution can be calculated using: $C = \frac{n}{V}$
where C is the concentration (in $\text{mol}\cdot\text{dm}^3$), n is the number of moles of solute dissolved in the solution and V is the volume of the solution (in dm^3).
- A **standard solution** is a solution in which the concentration is known to a high degree of precision. For the purposes of calculations, a standard solution can be thought of as one in which the concentration is a set value.
- A titration is a technique for determining the concentration of an unknown solution. We can calculate the unknown concentration using:

$$\frac{C_A V_A}{a} = \frac{C_B V_B}{b}$$

- A **limiting reagent** is a reagent that is completely used up in a chemical reaction.
- An **excess reagent** is a reagent that is not completely used up in a chemical reaction.
- **Percent yield** is calculated using:

$$\% \text{yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

Where the actual yield is the amount of product that is produced when you carry out the reaction and the theoretical yield is the amount of product that you calculate for the reaction using stoichiometric methods.

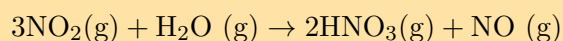
- The **empirical formula** is the simplest formula of a compound.
- The **molecular formula** is the full formula of a compound.
- **Percent purity** is calculated using:

$$\% \text{purity} = \frac{\text{mass of compound}}{\text{mass of sample}} \times 100$$

Physical Quantities		
Quantity	Unit name	Unit symbol
Concentration (C)	moles per cubic decimetre	$\text{mol}\cdot\text{dm}^{-3}$
Mass (m)	kilogram	kg
Molar mass (M)	gram per mole	$\text{g}\cdot\text{mol}^{-1}$
Moles (n)	moles	mol

Exercise 8 – 9:

- Write only the word/term for each of the following descriptions:
 - A reagent that is completely used up in a chemical reaction.
 - The simplest formula of a compound.
 - The amount of product that is calculated for a reaction using stoichiometric methods.
 - A technique for determining the concentration of an unknown solution.
- What is the volume of 3 mol of N_2 gas at STP?
 - 67,2 dm³
 - 22,4 dm³
 - 33,6 dm³
 - 63,2 dm³
- Given the following reaction:



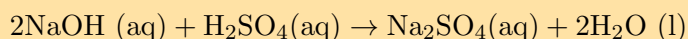
If 2,7 dm³ of NO_2 is used, what volume of HNO_3 is produced?

- 4,1 dm³
 - 2,7 dm³
 - 1,8 dm³
 - 3,4 dm³
- If 4,2 g of magnesium sulfate is dissolved in 350 cm³ of water, what is the concentration of the solution?
 - 0,1 mol·dm⁻³
 - 0,05 mol·dm⁻³
 - 0,003 mol·dm⁻³
 - 0,0001 mol·dm⁻³
 - Gold is occasionally found as the mineral calaverite. Calaverite contains a telluride of gold (AuTe_2). Phumza wants to know how much calaverite is in a sample of rock. She finds that the rock sample weighs 3,6 g. After performing some experiments she finds that the mass of calaverite and the crucible is 2,4 g. The mass of the crucible is 0,3 g. What is the percent purity of calaverite in the sample?
 - 63%
 - 54%
 - 58%
 - 67%
 - 300 cm³ of a 0,1 mol·dm⁻³ solution of sulfuric acid is added to 200 cm³ of a 0,5 mol·dm⁻³ solution of sodium hydroxide.

- Write down a balanced equation for the reaction which takes place when these two solutions are mixed.
- Calculate the number of moles of sulfuric acid which were added to the sodium hydroxide solution.
- Is the number of moles of sulfuric acid enough to fully neutralise the sodium hydroxide solution? Support your answer by showing all relevant calculations.

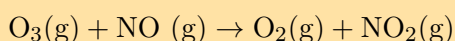
(IEB Paper 2 2004)

7. Given the equation:



25 cm³ of a 0,7 mol·dm⁻³ sulfuric acid (H₂SO₄) solution was pipetted into a conical flask and titrated with sodium hydroxide (NaOH). It was found that 23 cm³ of the sodium hydroxide was needed to neutralise the acid. Calculate the concentration of the sodium hydroxide.

8. Ozone (O₃) reacts with nitrogen monoxide gas (NO) to produce NO₂ gas. The NO gas forms largely as a result of emissions from the exhausts of motor vehicles and from certain jet planes. The NO₂ gas also contributes to the brown smog (smoke and fog), which is seen over most urban areas. This gas is also harmful to humans, as it causes breathing (respiratory) problems. The following equation indicates the reaction between ozone and nitrogen monoxide:



In one such reaction 0,74 g of O₃ reacts with 0,67 g NO.

- Calculate the number of moles of O₃ and of NO present at the start of the reaction.
- Identify the limiting reagent in the reaction and justify your answer.
- Calculate the mass of NO₂ produced from the reaction.

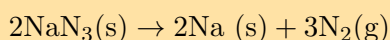
(DoE Exemplar Paper 2, 2007)

9. Calcium carbonate decomposes on heating to produce calcium oxide and oxygen according to the following equation:

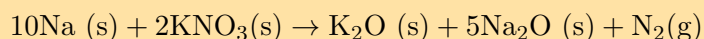


Thabang carries out the above reaction using 127 g of calcium carbonate. He finds that he gets 68,2 g of calcium oxide. What is the percentage yield?

10. Some airbags contain a mixture of sodium azide (NaN₃) and potassium nitrate (KNO₃). When a car crash is detected by the signalling system, the sodium azide is heated until it decomposes to form nitrogen gas and sodium metal:



The potassium nitrate then reacts with the sodium metal forming more nitrogen:



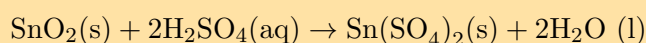
A typical passenger side airbag contains 250 g of sodium azide.

- a) What mass of sodium metal is formed in the first reaction?
- b) What is the total volume of nitrogen gas formed from both reactions?
- c) How much potassium nitrate (in g) is needed for all the sodium to be used up in the second reaction?

11. Chlorofluorocarbons (CFC's) are a class of compounds that have a long history of use in refrigerators. CFC's are slowly being phased out as they deplete the amount of ozone in the ozone layer. Jabu has a sample of a CFC that has the following percentage composition: 14,05% carbon, 41,48% chlorine and 44,46% fluorine.

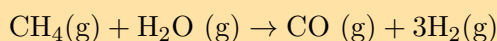
Determine the molecular formula of this CFC if the molar mass is $171 \text{ g}\cdot\text{mol}^{-1}$.

12. A sample containing tin dioxide (SnO_2) is to be tested to see how much tin dioxide it contains. The sample weighs 6,2 g. Sulfuric acid (H_2SO_4) is added to the sample and tin sulfate ($\text{Sn}(\text{SO}_4)_2$) forms. The equation for this reaction is:



If the mass of tin sulfate produced is 4,7 g, what is the percent purity of the sample?

13. Syngas (synthesis gas) is a mixture of carbon monoxide and hydrogen. Syngas can be produced from methane using:



Neels wants to make a mixture of syngas that has three times the volume of hydrogen gas.

- a) If the volume of methane used is 4 dm^3 , what volume of carbon monoxide and hydrogen will be produced?
- b) Will this amount of methane produce the correct mixture?

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